

TAO-DWS-001 — Dealer Website Scraping: Pre-fill Dealer Details Review

ISD Layer 1 — Dealer Website Scraping — Pre-fill Dealer Details Review (TAO-DWS-001)

```
1 template_version: ISD-v4.5-standard
2 layer: 1
3 isd_template: standard
4 orbit_version: v2.5
5 status: Draft
6 intent_id: TAO-DWS-001
7 intent_name: Dealer Website Scraping – Pre-fill Dealer Details Review
8 product_area: Dealer Onboarding / TAO
9 classification: Internal
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11 generated_by: paper-mcp isd_init (standard)
```

ISD Standard Template — Mid-tier template for moderate-complexity agents (A1-A2 autonomy). Includes all 13 base sections plus key v4.0 agent sections, but NOT enterprise process patterns (saga, batch, timers) or v4.2 completeness depth. For full enterprise coverage, use `template: "full"`. For non-AI features, use `template: "lite"`.

§1. Document Metadata

Field	Value
Intent ID	TAO-DWS-001
Intent Name	Dealer Website Scraping – Pre-fill Dealer Details Review
Product Area	Dealer Onboarding / TAO
Module / Sub-Area	TAO Configure Phase — Dealer Details Review Screen
Document Owner (PM)	pbihari@tekion.com
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Status	Draft

Version	1.0
Last Updated	Aug 27, 2026
Review Date	1st September
Source Reference	TAO_User_Journey.md — Dealer Details Review screen
Linked Persona(s)	Dealer Admin
ORBIT Template Version	ISD-v4.5-standard

1.2 Autonomy Declaration

Required for any intent that can execute without direct user invocation.
Skip for purely reactive intents.

Field	Value
Execution Model	hybrid — event-driven at Dealer Admin screen load AND proactive at IM provisioning stage
Autonomy Tier (ORBIT S04)	A1 (Suggest-Only) — scraper proposes prefilled values; human must confirm at every surface
Target Autonomy Tier (GA)	A1 (suggest-only) — confirmation always required; no auto-save of scraped data
Trigger Classification	event-triggered (Dealer Admin screen load) + proactive (IM provisioning stage)
Context Persistence	session-scoped for Dealer Admin path; provisioning-record-scoped for IM path (persisted per dealer record)
Execution Surface	TAO backend service — server-side; runs on both Dealer Admin screen load and IM provisioning stage
Background Execution	Yes — scrape runs in background; result cached; neither Dealer Admin screen nor IM

Autonomy Tier Reference (ORBIT S04):

- A0 (Tool): Deterministic function, no LLM reasoning
- A1 (Suggest-Only): Agent proposes action, human must approve every step
- A2 (Supervised): Agent executes routine actions, escalates exceptions to human

§2. Intent Definition

2.1 Primary Intent and Success Criteria

Field	Value
Primary Intent (business verb + scope)	Populate the TAO Dealer Details Review screen from two sources — (1) contract/Salesforce data as the always-present baseline, and (2) website scraping as an enrichment layer — then present a before/after comparison for Dealer Admin confirmation
Success Criteria	(1) Dealer Details Review screen is never blank — contract data always prefills it. (2) If a website exists, a comparison view shows what the website found vs. what is already on file, field by field. (3) Dealer Admin confirms the final values once; no manual re-typing of data already known to Tekion.
Value	High — eliminates blank-form experience for all dealers regardless of website presence; website enrichment layer adds data quality and signals intelligence
Frequency	Event-triggered — once per dealership at provisioning (contract sync) + once per Dealer Admin session on Dealer Details Review screen load
Feasibility	High — contract data is always available; website scraping is additive; no dealer is left

	with a blank form
Golden Intent?	Yes
Automation Potential	Partial — contract sync and scraping are fully automated; Dealer Admin confirmation of final values is always manual (A1)
Expected Volume	1 contract-sync invocation per dealer provisioning + 1 scrape per Dealer Admin session

2.2 User Personas

Persona	Role	Primary Use Case
Dealer Admin	Primary dealership-side owner; completes the TAO journey end-to-end	Reviews, edits, and confirms prefilled dealer details on the Dealer Details Review screen; can provide a corrected website URL if the pre-existing one is wrong
Implementation Manager (IM)	Tekion-side operator who sees the dealer pipeline and provisions dealers	Views scraped dealer details during the provisioning stage to verify dealer info before the Dealer Admin journey begins

2.3 Business Context

The Dealer Details Review screen must never be blank. When a dealer signs a contract with Tekion, their basic details — name, address, phone, website URL — are captured in Salesforce as part of that contract. Salesforce is already integrated with **TOC (Tekion Operating Cloud)** — the internal operational system through which the dealer's instance is provisioned and configured — and syncs dealer data into TOC as part of the contracting and provisioning pipeline. **TAO connects to TOC only; TAO does not read from Salesforce directly.** TOC is therefore the single source of dealer truth for TAO.

Layer 1 — Internal data sync (baseline, always present): At provisioning stage, TAO reads dealer details from **TOC**. TAO writes this record into the TAO dealer record. When the Dealer Admin opens the Dealer Details Review screen, all fields are already populated from this

baseline. This applies to every dealer — large, small, with or without a website. The no-blank-field guarantee comes from this layer.

Data flow: Salesforce → TOC ← TAO — Salesforce pushes dealer data into TOC; TAO reads from TOC.

Layer 2 — Website scraping (enrichment, additive): For dealers that have a public website, TAO also scrapes that website and compares what it finds against the merged internal record already on file. Contract/TOC data may be from months ago and may have a slightly different address or missing website URL. The website scrape surfaces these differences without discarding what Tekion already knows.

The Dealer Admin experience — before/after comparison view:

- All fields pre-populated from the merged internal record (Salesforce + TOC); labelled “From your records on file”
- Where the website found something different or new, a side-by-side comparison: “On file: 123 Main St | Found on website: 123 Main Street, Suite 100 | Use website version?”
- Where the website found the same value → a quiet checkmark (no action needed)
- Where the website found nothing new → on-file value shown alone, no comparison row

The Dealer Admin confirms the final state once. They are not re-typing anything — they are reviewing and accepting/rejecting diffs.

Small dealers with no website: Contract data fills the screen completely. No comparison rows appear because there is nothing to compare. The experience is: fully prefilled form from contract, Dealer Admin reviews and confirms.

dealer-admin-corrected website URL: If the website URL in the contract is wrong, missing, or the Dealer Admin wants to try a different page, the Dealer Details Review screen offers a “Try a different website” option that allows a fresh scrape. The same URL validation rules apply to any dealer-admin-provided URL before its scraped data is shown.

§3. Natural Language Triggers

3.1 Example Triggers

Trigger Phrase	Confidence Signal	Context
Dealer Details Review screen load event (system trigger)	High	Automatically fires when Dealer Admin opens the

		Dealer Details Review screen in TAO
“Prefill my dealership info”	High	Dealer Admin explicitly requests prefill from dealer website
“Use my website to fill in these details”	High	Dealer Admin provides intent to use website as source
“Look up my dealership”	Medium	Dealer Admin requests lookup — could be website scrape or internal lookup
“Fill in my address from my website”	Medium	Dealer Admin references a specific field; scrape is one resolution path

3.2 Disambiguation Notes

This intent is event-triggered (screen load), not conversational in the primary path. In a conversational context, “look up my dealership” could refer to internal TAO data lookup, external directory lookup, or dealer website scraping. Resolve by checking if dealer has a website URL on file; if yes, default to website scrape. If the Dealer Admin says “look up my address” without website context, clarify whether to use their website or another source.

3.3 Slash Commands

Command	Description
/prefill-dealer-details	Manually trigger a website scrape to re-populate the Dealer Details Review screen (for cases where auto-prefill failed or Dealer Admin wants a refresh)

3.4 Event Triggers (Autonomous Execution)

Required if Execution Model is proactive, scheduled, or event-driven. Skip for purely reactive intents.

Trigger Type	Event Source	Condition	Debounce	Exclusion Window
IM provisioning stage load	TAO IM portal — dealer list / provisioning queue view	Dealer record enters provisioning state AND dealer_website_url is present in contract data	Max 1 scrape per dealer provisioning record (re-scraped only if Dealer Admin provides a new URL)	None
Dealer Admin screen-load event	TAO frontend — Dealer Details Review screen	Screen renders for the first time in this session AND dealer_website_url is available	Max 1 auto-scrape per session; cached result reused on revisit	None
Dealer Admin provides alternate URL	Dealer Admin inputs a new URL on the Dealer Details Review screen	Dealer Admin submits a URL via “Use a different website” field AND URL passes validation (DWS-SI-08)	Max 1 per 5 minutes per dealership	None
Manual re-scrape (/prefill-dealer-details)	dealer-admin-invoked	Dealer Admin explicitly requests re-scrape with the current URL	Max 1 per 5 minutes per dealership	None

§4. Parameters

Parameter	Type	Required	Source	If Missing

dealership_id	string	Required	TAO session context	Block execution
session_id	string	Required	TAO session context	Block execution — required for caching
toc_dealer_data	object	Required	TOC dealer record — single source for TAO; holds Salesforce-synced contract data plus provisioning updates	Block execution — screen cannot render without TOC baseline
dealer_website_url	string (URL)	Optional	TOC dealer record (Dealer Website field, sourced from Salesforce contract at deal close)	Skip scraping layer; screen still renders from toc_dealer_data
dealer_provided_url	string (URL)	Optional	Dealer Admin input on Dealer Details Review screen (“Try a different website”)	N/A — only present when Dealer Admin explicitly provides an alternate URL; must pass DWS-SI-08 validation
scrape_timeout_ms	number	Optional	TAO backend config (default: 5000ms)	Use default 5000ms
fields_to_compare	string[] (enum)	Optional	TAO configuration	Default: dealership_name, dba_name, address lines,

				city, state, zip, phones, email, operating hours, and logo_primary; see §4.1
trigger_surface	enum: “im_provisioning” / “da_screen”	Required	System context	Default to “da_screen”

4.1 Field Mapping — TOC → Dealer Master API → Scrape Target

Every field in `fields_to_compare` maps to a specific path in the **Dealer Master API** response (`/dealer-master/{dealerId}`) and, where available, a **TOC source field** used for Layer 1 prefill. Fields without a TOC source are scrape-only and appear as “New — found on website” rows when the scraper finds them.

`website_url` is **NOT** in `fields_to_compare` — it is the trigger URL for scraping, not a field to compare.

4.1.1 Scrapeable Fields

#	<code>fields_to_compare</code> Key	Dealer Config UI Label	Dealer Master API Path	TOC Source Field	Layer	Scrape Target on Website
1	<code>dealership_name</code>	Dealer Name (Public)	<code>data.dealerName</code>	<code>Dealership Name</code>	1 + 2	Business name in header, footer, About page
2	<code>dba_name</code>	Doing Business As (DBA)	<code>data.dealerDoingBusinessAsName</code>	—	2 only	Footer “doing business as” / trading-name text
3	<code>address_line1</code>	Street Address 1	<code>data.dealerAddress1</code>	<code>Primary Physical</code>	1 + 2	Contact page /

		— Default location	ss[addressType=DEFAULT].streetAddress1	Address		footer address block
4	address_line2	Street Address 2 — Default location	data.dealerAddressss[addressType=DEFAULT].streetAddress2	—	2 only	Second address line when present
5	city	City — Default location	data.dealerAddressss[addressType=DEFAULT].city	City	1 + 2	Contact page / footer address block
6	state	State — Default location	data.dealerAddressss[addressType=DEFAULT].state	State	1 + 2	Contact page / footer address block
7	zip	ZIP Code — Default location	data.dealerAddressss[addressType=DEFAULT].zipCode	Zip Code	1 + 2	Contact page / footer address block
8	phone_main	Phone — Default location	data.dealerAddressss[addressType=DEFAULT].	Dealer Main Number	1 + 2	Contact page / footer / click-to-call element

			phoneNum ber			
9	phone_sales	Phone — Sales department	data.dealerAddress[addressType=SALES].phoneNumber	—	2 only	Department s section / Contact page Sales row
10	phone_service	Phone — Service department	data.dealerAddress[addressType=SERVICE].phoneNumber	—	2 only	Department s section / Contact page Service row
11	phone_parts	Phone — Parts department	data.dealerAddress[addressType=PARTS].phoneNumber	—	2 only	Department s section / Contact page Parts row
12	email	Email — Default location	data.dealerAddress[addressType=DEFAULT].email	—	2 only	Contact page, if published
13	operating_hours_sales	Sales Operating Hours (Mon–Sun)	data.dealerOperationSchedule[dealerOperationType	—	2 only	Hours page / Contact page Sales row

			<code>=1].businessHours[]</code>			
14	<code>operating_hours_service</code>	Service Operating Hours (Mon–Sun)	<code>data.dealerOperationSchedule[dealerOperationType=0].businessHours[]</code>	—	2 only	Hours page / Contact page Service row
15	<code>operating_hours_parts</code>	Parts Operating Hours (Mon–Sun)	<code>data.dealerOperationSchedule[dealerOperationType=2].businessHours[]</code>	—	2 only	Hours page / Contact page Parts row
16	<code>logo_primary</code>	Dealership Banner Logo	<code>data.dealerLogos[title=DEALERSHIP_BANNER_IMAGE].imageUrl</code>	—	2 only	Logo image in site header — highest-resolution image detected

Layer column: `1 + 2` = TOC prefill baseline exists AND website scrape may update it. `2 only` = no TOC source; field appears only when scraper finds it.

Operating hours schema: each `businessHours[]` entry is `{dayOfWeek: "MONDAY"... "SUNDAY", openingTime: "HH:mm", closingTime: "HH:mm", closed: bool}`. Scraper returns the same shape; comparator diffs day-by-day.

4.1.2 Fields NOT Scrapeable — Must Be Collected from Dealer or OEM

Dealer Config UI Label	Dealer Master API Path	Collection Method
OEM Dealer ID	<code>data.oemDealerId</code>	Collect from dealer / OEM portal
Federal EIN / Tax ID	<code>data.einConfiguration.federalTaxId</code>	Collect from dealer — not public
Bank Details	<code>data.bankDetails[]</code>	Collect from dealer — ACH/banking setup
OEM + Makes (identity)	<code>data.oemDetails[].oem</code> , <code>data.supportedMakes[]</code>	From TOC; website can confirm but is not the source

§5. Execution Flow

5.1 Sub-Intents

Sub-Intent	Description	Parameters / Notes	Trigger Condition	Safety Gate	Phase
DWS-SI-00: Sync TOC dealer data to TAO dealer record	At IM provisioning stage, read dealer details from TOC and write the TOC dealer record into the TAO dealer record.	dealership_id, toc_dealer_data, trigger_surface="im_provisioning"	Dealer enters provisioning state in TAO	Required — blocks Dealer Admin screen load if not complete	MVP
DWS-SI-00b: IM provisioning scrape preview	After TOC sync, proactively scrape dealer website and surface a comparison preview in IM	dealership_id, dealer_website_url	DWS-SI-00 completes AND dealer_website_url is present	IM preview is read-only; IM cannot confirm or save	MVP

	dealer list view.				
DWS-SI-01a: Load internal data prefill	On Dealer Admin screen load, read the TOC-sourced dealer record and render it as the baseline prefill.	dealership_id, session_id	Dealer Admin screen loads	Required — screen must never be blank	MVP
DWS-SI-01b: Resolve dealer website URL	Look up dealer website URL from contract data; check for IM pre-scrape cache.	dealership_id, session_id	DWS-SI-01a completes	If no URL → skip to DWS-SI-04	MVP
DWS-SI-02: Scrape public website	Fetch and parse dealer website homepage + contact/about page; read-only GET only.	dealer_website_url OR validated dealer_provided_url, scrape_timeout_ms	URL resolved or alternate URL validated	Rate limit; 5s timeout; robots.txt; read-only GET; silent fallback	MVP
DWS-SI-03: Compare scraped data against contract data	Classify each field as Match, Different, New, or Missing. Operating hours are compared day-by-day; logo is	raw_scraped_html, dealer_master_api_field_map	Scrape completed	Only scraped fields >= 0.80 confidence enter comparison	MVP

	compared as URL presence only.				
DWS-SI-04: Present comparison view to Dealer Admin	Render baseline fields, side-by-side diff/new rows, quiet checkmarks for matches, and baseline-only fields for missing values.	structured_comparison_map, confidence_scores	DWS-SI-03 complete OR no scrape ran	A1 gate — Dealer Admin must submit; no value saved without explicit action	MVP
DWS-SI-05: No-scrape path	If dealer has no website URL, scrape fails, or times out, Dealer Admin sees contract-prefilled screen with no comparison rows.	error_type, fallback_mode	No URL found OR DWS-SI-02 fails/times out	No degradation; Dealer Admin confirms what is there	MVP
DWS-SI-06: Save confirmed dealer details	Persist Dealer Admin-confirmed final values to TAO dealer record.	confirmed_fields, dealership_id, session_id	Dealer Admin submits confirmed details	Required field validation before save	MVP
DWS-SI-07: Dealer Admin	Dealer Admin inputs a	dealer_provided_url	Dealer Admin submits a URL	Route to DWS-SI-08	MVP

provides alternate website URL	corrected or alternate URL via “Try a different website”.				
DWS-SI-08: Validate user-provided URL	Check URL format, domain block-list, and fuzzy-match extracted business name against contract dealership name.	dealer_provided_url, dealership_id	DWS-SI-07 fires	Hard gate with rejection or confirmation prompt	MVP

§6. Capabilities, Skills & Agent Assignment

6.1 Agent Assignment

Field	Value
Agent	TAO Onboarding Agent (T1 Base)
Agent Description	Handles dealer-side self-service onboarding steps within TAO; this intent adds a web scraping sub-capability to the Configure Phase screen load sequence

6.2 Capability Needs

Capability Need	Business Description	Data Returned
Salesforce contract data read	Read dealer details from Salesforce contract record — captured at contract signature	Structured dealer details object

TOC dealer data read	Read dealer details from TOC dealer record — operational details added or updated during provisioning setup	Structured dealer details object
TAO dealer record write (internal sync)	Write the TOC-sourced record into the TAO dealer record at provisioning	Write confirmation
Public website fetch	Retrieve HTML content from the dealer's public website URL	Raw HTML
HTML text extraction + field compare	Parse HTML, extract structured text fields, assign confidence scores, and compare against on-file contract data	Structured comparison map
Comparison view render	Render the before/after comparison UI on the Dealer Details Review screen	Per-field comparison rows shown to Dealer Admin
TAO dealer record write (confirmed)	Persist dealer-admin-confirmed final values after Dealer Admin submits	Confirmation receipt

6.3 Skill Inventory

Skill Name	Knowledge Description	Capabilities Used	Reusable Across
dealer-web-scrape	Knows how to fetch a dealer's public website, identify contact/about sections, and extract business name, address, phone, and URL fields	Public website fetch, HTML text extraction	Any TAO step that needs dealer info from public website

dealer-details-mapper	Maps raw scraped text to the Dealer Master API field schema; handles address normalization, phone formatting, operating hours parsing, logo URL extraction, and confidence scoring	HTML text extraction + field comparison	Future TAO screens and Dealer Master API enrichment flows
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6.4 Data Sources

Data Source	Description	Access Pattern
TOC (Tekion Operating Cloud) dealer record	Single source of dealer truth for TAO. Holds Salesforce-synced contract data plus provisioning-stage updates.	Read via TOC API at provisioning time; TAO does not connect to Salesforce directly
TAO dealer record	TAO internal dealer data store; holds the TOC-sourced record used to prefill the Dealer Admin screen	Read at Dealer Admin screen load; written at provisioning sync and after Dealer Admin confirmation
Dealer's public website	Publicly accessible HTML — enrichment/comparison layer	Read-only HTTP GET; respects robots.txt; no auth; homepage + contact/about page

6.5 Content Requirements

Content Type	Audience	Format	Notes
Internal-data prefill baseline	Dealer Admin	All fields pre-populated from merged Salesforce+TOC record; labelled “From your records on file”	Always present; never blank

Comparison diff row	Dealer Admin	Per-field side-by-side row: “On file / Found on your website / Use website version?”	Only shown for different or new website values
Match indicator	Dealer Admin	Quiet green checkmark: “✓ Matches your website”	Shown when website confirms the on-file value
IM comparison preview	Implementation Manager	Read-only inline row in dealer list	Informational only
URL validation rejection	Dealer Admin	Inline error explaining the provided website does not appear to be for the dealership	Shown on failed business-name fuzzy match
Borderline match confirmation prompt	Dealer Admin	“We found [scraped name] on this website — is this your dealership?”	Shown for fuzzy match score 0.70–0.85

§7. Business Rules & Safety

7.1 ECI Contract — Action Safety Levels

Sub-Intent	Risk Level	Autonomous Above (confidence)	Requires Confirmation	Notes
DWS-SI-00: Internal data sync to TAO record	write (internal)	1.0	false	IM-triggered; reads TOC; no dealer involvement
DWS-SI-00b: IM provisioning scrape preview	read-only	1.0	false	IM sees comparison preview; informational only

DWS-SI-01a: Load contract prefill	read-only	1.0	false	Required baseline; never skipped
DWS-SI-02: Scrape public website	read-only	1.0	false	Read-only GET only
DWS-SI-03: Compare scraped vs. contract data	read-only	0.80	false	Scraped fields below 0.80 suppressed
DWS-SI-04: Present comparison view	read-only (display)	N/A	true	A1 gate: no value saved without Dealer Admin submit
DWS-SI-05: No- scrape path	read-only (display)	N/A	true	Contract data already shown
DWS-SI-06: Save confirmed details	write	N/A	true	Dealer Admin submit is the trigger
DWS-SI-08: Validate user- provided URL	read-only (validation)	N/A	true	Hard gate

7.2 Business Rules — Policy Lake Entries

Policy ID	Rule	Trigger	Dealer- Configurable?	Scope
POL-TAO-DWS- 001-000	The Dealer Details Review screen must never be blank: TOC dealer data must be synced before the Dealer	Every Dealer Admin screen load	No	DWS-SI-00, DWS-SI-01a

	Admin can open this screen			
POL-TAO-DWS-001-000a	Data flow is Salesforce → TOC → TAO. TAO reads from TOC only; TOC owns reconciliation.	DWS-SI-00	No	DWS-SI-00
POL-TAO-DWS-001-001	Scraping is read-only: HTTP GET only — never POST, submit forms, authenticate, or store cookies	Every scrape attempt	No	All scrape executions
POL-TAO-DWS-001-002	Dealer Admin must explicitly confirm final values before data is saved	Every Dealer Admin session	No	DWS-SI-04, DWS-SI-05, DWS-SI-06
POL-TAO-DWS-001-003	No-website is not a degraded state; no website-related message appears when no URL exists	Dealer has no website URL	No	DWS-SI-05
POL-TAO-DWS-001-004	Scrape failure is not a degraded state; fallback silently to contract-prefilled form	Scrape failure or timeout	No	DWS-SI-05
POL-TAO-DWS-001-005	Only fields with confidence >=	Post field-mapping	No	DWS-SI-03, DWS-SI-04

	0.80 may be shown as prefilled without warning			
POL-TAO-DWS-001-006	Scraping must respect robots.txt; if disallowed, skip the scrape	Pre-scrape robots.txt check	No	DWS-SI-02
POL-TAO-DWS-001-007	User-provided URLs must pass domain validation before scraping	DWS-SI-08	No	DWS-SI-08
POL-TAO-DWS-001-008	User-provided URL result must fuzzy-match known dealership name at ≥ 0.70 similarity	DWS-SI-08 post-scrape validation	No	DWS-SI-08
POL-TAO-DWS-001-009	IM pre-scraped data is informational only; only Dealer Admin can confirm dealer details	IM provisioning stage	No	DWS-SI-00

7.3 Escalation Rules

Condition	Escalation Path	Notification Target
Scrape consistently fails for a dealership across 3+ sessions	Log to TAO ops dashboard for review; do not alert Dealer Admin	TAO ops team (internal only)

Dealer Admin manually overrides all prefilled fields in 3+ consecutive sessions	Flag pattern to PM team for review	TAO product analytics
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7.4 Compliance & Legal Constraints

Web scraping a publicly accessible website generally does not raise legal issues when done for reading the business's own published public information. However:

- TAO must respect `robots.txt` directives on the dealer's website.
- Only the dealer's own website URL may be scraped — no third-party lookups for MVP.
- No personal data is scraped; all scraped fields are business-level public details.
- Legal review recommended before GA if scope expands beyond MVP fields.

7.5 Data Access Constraints

Required per ORBIT S26 I227. Specify RBAC rules, PII handling, and cross-dealer isolation.

Constraint Type	Rule	Applies To	Notes
RBAC	Dealer Admin can view and confirm prefilled details and provide alternate URL. IM can view read-only scraped preview and cannot confirm or save.	DWS-SI-00, DWS-SI-04, DWS-SI-06	Module Champions and PSMs do not interact with either surface
PII Handling	Scraped fields are business-level public details, not personal data.	All sub-intents	If future fields include individual contact names, PII policy must be revisited
Cross-Dealer Isolation	Scrape cache is keyed by (dealership_id, session_id) — no cross-dealer data leakage	DWS-SI-01 through DWS-SI-06	

Data Residency	Scraped data cached and stored in US region only for MVP	DWS-SI-02, DWS-SI-03	Align with TAO's existing data residency policy
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7.6 Known Risks

Risk ID	Description	Likelihood	Impact	Mitigation	Owner
R-001	Dealer website is not publicly accessible, behind auth, or has no contact page	High	Low	Silent fallback; no error shown to Dealer Admin	Engineering
R-002	Scraped values are wrong or outdated	Medium	Medium	Confidence gate; Dealer Admin always confirms	PM + UX
R-003	Dealer website blocks scraping via robots.txt	Medium	Low	robots.txt check before scraping; silent skip	Engineering
R-004	Scrape is too slow (>5s) and blocks screen load	Low	High	Hard 5s timeout; async execution	Engineering
R-005	Dealer has no website at all	High for small dealers	None	No-website is a first-class fallback	PM + Engineering
R-006	Dealer Admin provides an intentionally wrong URL	Low to Medium	High	Domain block-list + business-	Engineering + PM

				name fuzzy match	
R-007	Dealer Admin provides group HQ site instead of rooftop site	Medium	Medium	Borderline match prompt for 0.70–0.85	Engineering

7.10 Financial Controls

Required for financial process intents. Define tolerance bands, 3-way match rules, GL coding, reconciliation gates, segregation of duties, and audit trail requirements. Skip if not applicable.

N/A — this intent does not handle financial transactions. Dealer website scraping is a data pre-fill feature; no monetary values, GL coding, or financial reconciliation are involved.

§8. Rendering Declaration

8.1 Rendering Mode

Sub-Intent / Response Type	Preferred Mode	Fallback	Required Components
Prefilled fields on Dealer Details Review screen	agui (inline form with pre-populated fields, editable)	basic	Prefill indicator label; confidence badge; editable inputs; confirm/submit button
Scrape failure message	basic (inline advisory text, non-blocking)	text	“We couldn't find your details automatically — please fill in below” label
Manual re-scrape trigger (/prefill-dealer-details)	basic (status feedback)	text	Inline spinner or progress indicator; update fields when complete

8.2 Orchestration Handoffs

When this intent hands control to another agent mid-flow, declare the handoff contract here.

Handoff Trigger	Target Agent / Intent	Context Passed	Return Expected	Handoff Type	Notes
Dealer Admin confirms dealer details	TAO Configure Phase — next step in journey	Confirmed dealer detail fields, dealership_id, session_id	Yes — proceed to next TAO step	Blocking	Standard TAO step-completion handoff

§9. Conversation Design & Test Scenarios

9.1 Happy Path Scenarios

#	Trigger	Agent Does	Screen Result
1	IM opens provisioning queue; dealer has website URL in contract	DWS-SI-00 → DWS-SI-00b → builds comparison map	IM sees comparison preview inline with matches and diffs
2	Dealer Admin opens screen; scrape result cached from IM pre-scrape	DWS-SI-01a → DWS-SI-03 → DWS-SI-04	Dealer Admin sees prefilled fields, diff rows, and confirms final values
3	Dealer Admin opens screen; small dealer, no website URL	DWS-SI-01a → DWS-SI-01b → DWS-SI-05	All fields populated from contract; no comparison rows; no mention of websites
4	Website scrape finds same values as contract on all fields	DWS-SI-03 classifies all fields as Match	Quiet “✓ Matches your website” indicators; no comparison rows

5	Dealer Admin provides alternate URL with borderline fuzzy match	DWS-SI-08 shows confirmation prompt	Dealer Admin confirms whether the found website belongs to their dealership
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9.2 Edge Case Scenarios

#	Trigger	Condition	Agent Does	Screen Result
1	Dealer Admin opens screen	No website	DWS-SI-01b finds no URL → contract-only view	All fields pre-populated from contract; no comparison rows
2	Dealer Admin opens screen	Website returns 403 / behind auth	Scrape fails → DWS-SI-05	Same as no-website case
3	Website scrape runs	Phone confidence 0.62	DWS-SI-03 suppresses phone	Phone comparison row does not appear
4	Scrape times out	5s timeout	DWS-SI-05	Screen loads immediately from contract data; no spinner blocking Dealer Admin
5	Dealer Admin provides alternate URL	Domain on block-list	DWS-SI-08 rejects immediately	Shows “This doesn't look like a dealership website.”
6	Dealer Admin provides wrong dealership URL	Business name mismatch	DWS-SI-08 rejects post-scrape	Shows “The website you provided doesn't appear to be for [Dealership Name].”
7	Dealer Admin opens screen	Contract sync has not	DWS-SI-01a cannot load	Screen shows loading state;

		completed	contract data	Dealer Admin cannot proceed
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9.3 Edge Case Catalog

ID	Condition	Expected Behavior	Severity	Test Type
EC-001	Contract sync not complete	Show loading state; block Dealer Admin until sync completes	Blocking	Dependency
EC-002	dealer_website_url is null	Skip scraping layer; render contract-only view	Non-blocking	Negative
EC-003	Dealer website behind auth or scrape fails	Scraping skipped; contract-only view shown	Non-blocking	Boundary
EC-004	robots.txt disallows scraping	Skip scrape silently	Safety Gate	Security
EC-005	All scraped fields match contract data	Suppress comparison rows; show quiet match indicators	Non-blocking	Happy path variant
EC-006	User-provided URL on domain block-list	Reject immediately	Blocking	Security
EC-007	User-provided URL similarity < 0.70	Reject with mismatch message	Blocking	Data quality

EC-008	Borderline match (0.70–0.85)	Show confirmation prompt	Warning	Data quality
EC-009	Dealer Admin submits with required fields empty	Block submit; required field validation error	Blocking	Validation
EC-010	Dealer Admin re-triggers scrape within 5 minutes	Reject with “Please wait before retrying”	Non-blocking	Rate limit

9.4 Golden Test Cases

Minimum 5 golden test cases per intent (Agentic Testing Principle 3).
 Sourced from real production scenarios.

GTC#	Input	Expected Output	Quality Dimensions	Pass Criteria
GTC-001	Dealer with website; address differs from contract and website URL is found	Address diff row and “New from website” URL row shown; Dealer Admin accepts desired values	Accuracy; Completeness	Correct diffs surfaced; no false-positive diff rows
GTC-002	Small dealer; dealer_website_url=null	All fields pre-populated from contract; no comparison rows	No-website baseline	Screen fully prefilled; zero website UI
GTC-003	Website response time 6s	Screen loads from contract data immediately; scrape times out silently	Timeout enforcement	Screen visible in < 500ms; no error

GTC-004	Dealer Admin provides blocked domain	DWS-SI-08 rejects immediately; comparison view unchanged	Domain validation	Immediate rejection; contract prefill unaffected
GTC-005	Dealer Admin provides competitor URL	DWS-SI-08 rejects post-scrape; contract prefill unaffected	Business-name fuzzy match	Rejection message shown
GTC-006	trigger_surface="im_provisioning"; address diff found	IM dealer list row shows contract values + address diff highlighted	IM preview	IM sees diff inline; no IM action/save possible

9.5 Quality Dimensions

Explicit quality dimensions with thresholds for every LLM-generated output (Principle 2).

Dimension	Description	Target Threshold	Regression Threshold	Eval Method
Accuracy	Factual correctness of responses	>= 95%	< 90% blocks	Deterministic + LLM-as-judge
Completeness	All required information included	>= 90%	< 85% blocks	Schema validation
Tone	Professional, contextually appropriate	>= 4/5	< 3/5 flags	LLM-as-judge
Safety	No harmful, private, or hallucinated content	100%	< 100% blocks	Deterministic + red-team

9.6 Failure Mode Coverage

Failure Type	Applicable?	Specific Risk	Mitigation	Test Strategy
Logic	Y	Field mapping may misidentify fields on non-standard dealer websites	Confidence scoring + field-type validation	Unit tests with diverse HTML fixtures
Integration	Y	Website unreachable, slow, redirecting, or unexpected content type	Timeout, robots.txt check, MIME type guard	Pipeline tests with mock HTTP responses
Behavioral	N	No LLM reasoning in happy path	N/A	N/A
Quality	Y	Scraped values may be outdated or incorrect	Confidence threshold; Dealer Admin always confirms	Golden set with known-correct vs. outdated websites
Drift	Y	Dealer website HTML structure changes over time	Monitor scrape success rates	Weekly regression on dealer URL golden set
Context	N	Intent scoped to Dealer Details Review screen	N/A	N/A
Safety	Y	Scrape must not write to, submit forms on, or authenticate with dealer website	Read-only HTTP GET; no cookies/session storage	Red-team verification that no POST/PUT/auth calls are made

Coordination	N	Intent operates independently	N/A	N/A
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§10. Success Metrics & Evaluation

10.1 Business KPIs

KPI	Current Baseline	Target	Measurement Method	Timeline
Blank-field rate at screen open	~100% (all fields blank today)	0%	TAO analytics — blank screen loads / total loads	At TAO V1 launch
Dealer Details screen completion time	TBD — measure at TAO V1 launch	Reduce by >= 40% vs. baseline	TAO analytics — screen open to submit	30 days post-launch
Dealer Admin field correction rate	N/A	< 15% of contract-prefilled fields manually corrected	DA_edit events / total contract fields shown	30 days post-launch
Website diff acceptance rate	N/A	Target >= 50%	Accepted website diffs / total diff rows shown	30 days post-launch
Dealer Details form validation error rate	TBD — measure at TAO V1 launch	Reduce by >= 60% vs. baseline	TAO analytics — validation errors on submit	30 days post-launch

10.2 Agent Evaluation Criteria

Metric	Definition	Target	Regression Threshold	Cadence
Intent match rate	% correctly classified	> 95%	< 90% triggers review	Weekly

Golden set pass rate	% golden scenarios passing	> 98%	< 95% blocks release	Per deploy
Quality dimension scores	Per-dimension eval scores	Per §9.5 targets	Per §9.5 regression	Per deploy
Invariant violation rate	% operations violating invariants	0%	> 0% blocks release	Continuous

10.3 Invariants (Continuous Checks)

Properties that must ALWAYS hold — checked on every operation, not periodically (Principle 5).

Invariant	Description	Check Frequency	Enforcement
Layer 1 must precede Layer 2	Dealer Admin screen must never render without Layer 1 contract data present	Every screen load	Hard block
Screen is never blank	Contract values fill every field; Layer 2 cannot make a field emptier than Layer 1 left it	Every render	Hard block
Read-only scrape	Scraper never submits forms, writes cookies, or authenticates	Every scrape	Hard block
Dealer Admin confirmation required before save	No value persisted without explicit Dealer Admin submit action	Every save	Hard block

Dealer isolation	Scrape/comparison results for one dealership never shown to another	Every render	Hard block
URL validation before user-provided scrape	DWS-SI-02 must never run on dealer_provided_url unless DWS-SI-08 passed it	Every relevant scrape	Hard block
IM cannot save dealer details	IM surface is read-only; write endpoint rejects IM provisioning saves	Every write	Hard block
Data integrity	No orphaned references, append-only IDs	Every write	Hard block
Privacy	No private content in external outputs	Every export	Hard block

10.4 Related Intents & Dependencies

Upstream Dependencies

Intent ID	Intent Name	Dependency Type	Notes
TAO-CONTRACT-001 (implicit)	Contract Signature & Dealer Intake	Data	dealer_website_url must be captured at contract signature
TAO-COA-001	COA / Financial Statement Upload (Step 7)	Workflow	Same Configure Phase; no direct dependency

Downstream Effects

Intent ID	Intent Name	Effect Type	Notes

TAO Configure Phase — next step	Dealer Details confirmed → Configure Phase proceeds	Workflow	DWS-001 completion unblocks the next step
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§11. Follow-Up Patterns

Pattern	Trigger	Agent Response
Dealer Admin edits a prefilled field	Dealer Admin changes a prefilled value in the form	Accept the edit silently; edited value replaces prefilled value — Dealer Admin is always the source of truth
Dealer Admin requests re-scrape	Dealer Admin triggers /prefill-dealer-details	Check debounce window; if allowed, re-run DWS-SI-02 through DWS-SI-04; if within debounce, show “Please wait before retrying”
Prefill partially succeeds	DWS-SI-03 returns a partial field map	Prefill found fields; leave unfound fields blank with standard placeholder text; do not show partial-failure message

§12. Technical Approach

Standard feature mode — engineering-review format. AI capability mapping not required.

Architecture — Two-Layer Model

This feature has two independent layers that compose on the Dealer Details Review screen.

Layer 1 — Internal Data Sync (baseline, always runs)

Data flow: Salesforce → T0C → TAO

1. **TOC Sync Service** — At IM provisioning stage, reads the dealer record from TOC. TAO does not connect to Salesforce directly — TOC is the single API call.
 2. **TAO Dealer Record** — Stores the TOC-sourced dealer record. Dealer Admin screen reads from here. Always populated before the Dealer Admin journey begins.
-

Layer 2 — Website Scraping + Comparison (enrichment, additive)

3. **Scrape Trigger** — Resolves the dealer website URL from the TAO dealer record. If no URL, Layer 2 is skipped entirely.
 4. **Web Scraper Worker** — HTTP GET to target URL (homepage + contact/about page). Enforces 5s timeout, robots.txt, MIME type guard, and read-only GET.
 5. **Field Extractor** — Parses raw HTML and extracts dealership name, DBA, address, phones, email, operating hours, and logo URL with confidence scores.
 6. **Field Comparator** — Compares extracted values against contract values and classifies fields as match, different, new, or missing.
 7. **Comparison View Renderer** — Injects comparison map into the Dealer Details Review screen.
 8. **Session Cache** — Caches comparison result keyed by (dealership_id, session_id).
 9. **TAO Dealer Record Write** — On Dealer Admin submit, confirmed final values are persisted.
-

URL Validation (for dealer-admin-provided alternate URLs)

10. **URL Validator** — Domain block-list check; scrape and fuzzy-match extracted business name against contract dealership name. Below 0.70 rejects, 0.70–0.85 prompts, and ≥ 0.85 accepts silently.
-

Non-Functional Requirements

- **Screen load latency:** Dealer Admin screen must render from Layer 1 data in $< 500\text{ms}$.
 - **Layer 1 is blocking:** Dealer Admin cannot open the Dealer Details Review screen until DWS-SI-00 is complete.
 - **Layer 2 is always additive:** Any Layer 2 failure falls back silently to Layer 1 data only.
 - **Rate limiting:** Max 1 scrape per (dealership_id, session); max 1 user-triggered re-scrape per 5 minutes per dealership.
 - **Security:** Read-only HTTP GET only. No auth. No cookies. No form submissions.
-

§13. Rollout Plan

Standard feature mode — TAO V1 launch.

Rollout Target: TAO V1

This feature ships with TAO V1. There is no separate phased rollout — the feature is included in the initial product release. Rollout is controlled by feature flag `dws_prefill_enabled` for rapid kill-switch.

Gate	Criteria	Owner
Pre-launch	Zero P0/P1 bugs in golden test cases GTC-001 through GTC-006	Engineering
Pre-launch	DWS-SI-08 domain block-list reviewed and approved by PM + Legal	PM
Pre-launch	Business-name fuzzy match thresholds validated on real dealer names	Engineering + PM
Pre-launch	IM provisioning view updated to show scraped preview column	Engineering
Pre-launch	“Use a different website” UI component built and tested	Engineering + UX
Post-launch monitoring	Scrape success rate $\geq 50\%$; timeout rate $\leq 10\%$; Dealer Admin edit rate $< 25\%$	PM
Rollback trigger	P0: any scrape that writes to a dealer website; any prefilled value saved without Dealer Admin confirmation	Engineering on-call

Feature flag: `dws_prefill_enabled` — set to `false` disables all scraping immediately. No data migration needed for rollback.

Appendix A: ISD Versioning

Version	Date	Author	Change Summary
1.0	Aug 27, 2026	Prem Bihari (pbihari@tekion.com)	Initial ISD draft via paper-mcp isd_init (standard)

Appendix B: PM-002 Scoring Rubric — Standard Mode (Reference)

ISD Standard uses the same 13-section base rubric as the full template (100 base points) plus an 8-point v4.0 autonomy bonus for the two included v4.0 sections.

Section	Weight	Full Credit (100%)	Partial Credit (60%)	No Credit (0%)
§1 Metadata	5pts	All fields complete	Missing optional fields	Missing required fields
§2 Intent Definition	10pts	All criteria defined	Success criteria only	No meaningful definition
§3 NL Triggers	8pts	>= 5 triggers, mix, disambiguation	3-4 triggers	< 3 triggers
§4 Parameters	7pts	All params with source and fallback	Sources defined	Parameter list only
§5.1 Sub-Intents	15pts	IMS Recon granularity; all columns	Happy path only	< 5 sub-intents
§6 Capabilities/Skills	8pts	Business- language caps; agent clear	Caps listed, eng- language	Missing

§7 Business Rules	10pts	ECI contracts; policies identified; §7.5 present	ECI thresholds only	No safety model
§8 Rendering	5pts	Mode declared for all outputs	Primary output only	No declaration
§9 Conversations	8pts	Happy path + edge cases + catalog	Happy path only	No scenarios
§10 Metrics	8pts	KPIs + baselines + attribution	KPIs without baselines	No metrics
§11 Follow-up patterns	5pts	>= 2 patterns	1 pattern	None
§12 Technical Approach	6pts	Constraints + integrations + NFRs	Constraints only	Absent
§13 Rollout Plan	5pts	Phase gates + flags + rollback	Phase list	Absent

Autonomous Agent Bonus: §1.2 Autonomy Declaration (5pts) and §3.4 Event Triggers (3pts). Gate: Score >= 70 required for isd_export. Grand total: 108 max.

7.3 Escalation Rules

Condition	Escalation Path	Notification Target
Scrape consistently fails for a dealership across 3+ sessions	Log to TAO ops dashboard for review; do not alert Dealer Admin	TAO ops team (internal only)
Dealer Admin manually overrides all prefilled fields in 3+ consecutive sessions	Flag pattern to PM team for review	TAO product analytics

7.4 Compliance & Legal Constraints

Web scraping a publicly accessible website generally does not raise legal issues when done for reading the business's own published public information. However:

- TAO must respect `robots.txt` directives on the dealer's website.
- Only the dealer's own website URL may be scraped — no third-party lookups for MVP.
- No personal data is scraped; all scraped fields are business-level public details.
- Legal review recommended before GA if scope expands beyond MVP fields.

7.5 Data Access Constraints

Required per ORBIT S26 I227. Specify RBAC rules, PII handling, and cross-dealer isolation.

Constraint Type	Rule	Applies To	Notes
RBAC	Dealer Admin can view and confirm prefilled details and provide alternate URL. IM can view read-only scraped preview and cannot confirm or save.	DWS-SI-00, DWS-SI-04, DWS-SI-06	Module Champions and PSMs do not interact with either surface
PII Handling	Scraped fields are business-level public details, not personal data.	All sub-intents	If future fields include individual contact names, PII policy must be revisited
Cross-Dealer Isolation	Scrape cache is keyed by (dealership_id, session_id) — no cross-dealer data leakage	DWS-SI-01 through DWS-SI-06	
Data Residency	Scraped data cached and stored in US region only for MVP	DWS-SI-02, DWS-SI-03	Align with TAO's existing data residency policy

7.6 Known Risks

Risk ID	Description	Likelihood	Impact	Mitigation	Owner
R-001	Dealer website is not publicly accessible, behind auth, or has no contact page	High	Low	Silent fallback; no error shown to Dealer Admin	Engineering
R-002	Scraped values are wrong or outdated	Medium	Medium	Confidence gate; Dealer Admin always confirms	PM + UX
R-003	Dealer website blocks scraping via robots.txt	Medium	Low	robots.txt check before scraping; silent skip	Engineering
R-004	Scrape is too slow (>5s) and blocks screen load	Low	High	Hard 5s timeout; async execution	Engineering
R-005	Dealer has no website at all	High for small dealers	None	No-website is a first-class fallback	PM + Engineering
R-006	Dealer Admin provides an intentionally wrong URL	Low to Medium	High	Domain block-list + business-name fuzzy match	Engineering + PM

R-007	Dealer Admin provides group HQ site instead of rooftop site	Medium	Medium	Borderline match prompt for 0.70–0.85	Engineering
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7.10 Financial Controls

Required for financial process intents. Define tolerance bands, 3-way match rules, GL coding, reconciliation gates, segregation of duties, and audit trail requirements. Skip if not applicable.

N/A — this intent does not handle financial transactions. Dealer website scraping is a data pre-fill feature; no monetary values, GL coding, or financial reconciliation are involved.

§8. Rendering Declaration

8.1 Rendering Mode

Sub-Intent / Response Type	Preferred Mode	Fallback	Required Components
Prefilled fields on Dealer Details Review screen	agui (inline form with pre-populated fields, editable)	basic	Prefill indicator label; confidence badge; editable inputs; confirm/submit button
Scrape failure message	basic (inline advisory text, non-blocking)	text	“We couldn't find your details automatically — please fill in below” label
Manual re-scrape trigger (/prefill-dealer-details)	basic (status feedback)	text	Inline spinner or progress indicator; update fields when complete

8.2 Orchestration Handoffs

When this intent hands control to another agent mid-flow, declare the handoff contract here.

Handoff Trigger	Target Agent / Intent	Context Passed	Return Expected	Handoff Type	Notes
Dealer Admin confirms dealer details	TAO Configure Phase — next step in journey	Confirmed dealer detail fields, dealership_id, session_id	Yes — proceed to next TAO step	Blocking	Standard TAO step-completion handoff

§9. Conversation Design & Test Scenarios

9.1 Happy Path Scenarios

#	Trigger	Agent Does	Screen Result
1	IM opens provisioning queue; dealer has website URL in contract	DWS-SI-00 → DWS-SI-00b → builds comparison map	IM sees comparison preview inline with matches and diffs
2	Dealer Admin opens screen; scrape result cached from IM pre-scrape	DWS-SI-01a → DWS-SI-03 → DWS-SI-04	Dealer Admin sees prefilled fields, diff rows, and confirms final values
3	Dealer Admin opens screen; small dealer, no website URL	DWS-SI-01a → DWS-SI-01b → DWS-SI-05	All fields populated from contract; no comparison rows; no mention of websites
4	Website scrape finds same values as contract on all fields	DWS-SI-03 classifies all fields as Match	Quiet “✓ Matches your website” indicators; no comparison rows

5	Dealer Admin provides alternate URL with borderline fuzzy match	DWS-SI-08 shows confirmation prompt	Dealer Admin confirms whether the found website belongs to their dealership
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9.2 Edge Case Scenarios

#	Trigger	Condition	Agent Does	Screen Result
1	Dealer Admin opens screen	No website	DWS-SI-01b finds no URL → contract-only view	All fields pre-populated from contract; no comparison rows
2	Dealer Admin opens screen	Website returns 403 / behind auth	Scrape fails → DWS-SI-05	Same as no-website case
3	Website scrape runs	Phone confidence 0.62	DWS-SI-03 suppresses phone	Phone comparison row does not appear
4	Scrape times out	5s timeout	DWS-SI-05	Screen loads immediately from contract data; no spinner blocking Dealer Admin
5	Dealer Admin provides alternate URL	Domain on block-list	DWS-SI-08 rejects immediately	Shows “This doesn't look like a dealership website.”
6	Dealer Admin provides wrong dealership URL	Business name mismatch	DWS-SI-08 rejects post-scrape	Shows “The website you provided doesn't appear to be for [Dealership Name].”
7	Dealer Admin opens screen	Contract sync has not	DWS-SI-01a cannot load	Screen shows loading state;

		completed	contract data	Dealer Admin cannot proceed
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9.3 Edge Case Catalog

ID	Condition	Expected Behavior	Severity	Test Type
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EC-002	dealer_website_url is null	Skip scraping layer; render contract-only view	Non-blocking	Negative
EC-003	Dealer website behind auth or scrape fails	Scraping skipped; contract-only view shown	Non-blocking	Boundary
EC-004	robots.txt disallows scraping	Skip scrape silently	Safety Gate	Security
EC-005	All scraped fields match contract data	Suppress comparison rows; show quiet match indicators	Non-blocking	Happy path variant
EC-006	User-provided URL on domain block-list	Reject immediately	Blocking	Security
EC-007	User-provided URL similarity < 0.70	Reject with mismatch message	Blocking	Data quality

EC-008	Borderline match (0.70–0.85)	Show confirmation prompt	Warning	Data quality
EC-009	Dealer Admin submits with required fields empty	Block submit; required field validation error	Blocking	Validation
EC-010	Dealer Admin re-triggers scrape within 5 minutes	Reject with “Please wait before retrying”	Non-blocking	Rate limit

9.4 Golden Test Cases

Minimum 5 golden test cases per intent (Agentic Testing Principle 3).
Sourced from real production scenarios.

GTC#	Input	Expected Output	Quality Dimensions	Pass Criteria
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GTC-002	Small dealer; dealer_website_url=null	All fields pre-populated from contract; no comparison rows	No-website baseline	Screen fully prefilled; zero website UI
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GTC-005	Dealer Admin provides competitor URL	DWS-SI-08 rejects post-scrape; contract prefill unaffected	Business-name fuzzy match	Rejection message shown
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Explicit quality dimensions with thresholds for every LLM-generated output (Principle 2).

Dimension	Description	Target Threshold	Regression Threshold	Eval Method
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Behavioral	N	No LLM reasoning in happy path	N/A	N/A
Quality	Y	Scraped values may be outdated or incorrect	Confidence threshold; Dealer Admin always confirms	Golden set with known-correct vs. outdated websites
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Context	N	Intent scoped to Dealer Details Review screen	N/A	N/A
Safety	Y	Scrape must not write to, submit forms on, or authenticate with dealer website	Read-only HTTP GET; no cookies/session storage	Red-team verification that no POST/PUT/auth calls are made

Coordination	N	Intent operates independently	N/A	N/A
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§10. Success Metrics & Evaluation

10.1 Business KPIs

KPI	Current Baseline	Target	Measurement Method	Timeline
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Dealer Admin field correction rate	N/A	< 15% of contract-prefilled fields manually corrected	DA_edit events / total contract fields shown	30 days post-launch
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10.3 Invariants (Continuous Checks)

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Dealer Admin confirmation required before save	No value persisted without explicit Dealer Admin submit action	Every save	Hard block

Dealer isolation	Scrape/comparison results for one dealership never shown to another	Every render	Hard block
URL validation before user-provided scrape	DWS-SI-02 must never run on dealer_provided_url unless DWS-SI-08 passed it	Every relevant scrape	Hard block
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Data integrity	No orphaned references, append-only IDs	Every write	Hard block
Privacy	No private content in external outputs	Every export	Hard block

10.4 Related Intents & Dependencies

Upstream Dependencies

Intent ID	Intent Name	Dependency Type	Notes
TAO-CONTRACT-001 (implicit)	Contract Signature & Dealer Intake	Data	dealer_website_url must be captured at contract signature
TAO-COA-001	COA / Financial Statement Upload (Step 7)	Workflow	Same Configure Phase; no direct dependency

Downstream Effects

Intent ID	Intent Name	Effect Type	Notes

TAO Configure Phase — next step	Dealer Details confirmed → Configure Phase proceeds	Workflow	DWS-001 completion unblocks the next step
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§11. Follow-Up Patterns

Pattern	Trigger	Agent Response
Dealer Admin edits a prefilled field	Dealer Admin changes a prefilled value in the form	Accept the edit silently; edited value replaces prefilled value — Dealer Admin is always the source of truth
Dealer Admin requests re-scrape	Dealer Admin triggers /prefill-dealer-details	Check debounce window; if allowed, re-run DWS-SI-02 through DWS-SI-04; if within debounce, show “Please wait before retrying”
Prefill partially succeeds	DWS-SI-03 returns a partial field map	Prefill found fields; leave unfound fields blank with standard placeholder text; do not show partial-failure message

§12. Technical Approach

Standard feature mode — engineering-review format. AI capability mapping not required.

Architecture — Two-Layer Model

This feature has two independent layers that compose on the Dealer Details Review screen.

Layer 1 — Internal Data Sync (baseline, always runs)

Data flow: `Salesforce → TOC → TAO`

1. **TOC Sync Service** — At IM provisioning stage, reads the dealer record from TOC. TAO does not connect to Salesforce directly — TOC is the single API call.
 2. **TAO Dealer Record** — Stores the TOC-sourced dealer record. Dealer Admin screen reads from here. Always populated before the Dealer Admin journey begins.
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Layer 2 — Website Scraping + Comparison (enrichment, additive)

3. **Scrape Trigger** — Resolves the dealer website URL from the TAO dealer record. If no URL, Layer 2 is skipped entirely.
 4. **Web Scraper Worker** — HTTP GET to target URL (homepage + contact/about page). Enforces 5s timeout, robots.txt, MIME type guard, and read-only GET.
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 8. **Session Cache** — Caches comparison result keyed by (dealership_id, session_id).
 9. **TAO Dealer Record Write** — On Dealer Admin submit, confirmed final values are persisted.
-

URL Validation (for dealer-admin-provided alternate URLs)

10. **URL Validator** — Domain block-list check; scrape and fuzzy-match extracted business name against contract dealership name. Below 0.70 rejects, 0.70–0.85 prompts, and ≥ 0.85 accepts silently.
-

Non-Functional Requirements

- **Screen load latency:** Dealer Admin screen must render from Layer 1 data in $< 500\text{ms}$.
 - **Layer 1 is blocking:** Dealer Admin cannot open the Dealer Details Review screen until DWS-SI-00 is complete.
 - **Layer 2 is always additive:** Any Layer 2 failure falls back silently to Layer 1 data only.
 - **Rate limiting:** Max 1 scrape per (dealership_id, session); max 1 user-triggered re-scrape per 5 minutes per dealership.
 - **Security:** Read-only HTTP GET only. No auth. No cookies. No form submissions.
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§13. Rollout Plan

Standard feature mode — TAO V1 launch.

Rollout Target: TAO V1

This feature ships with TAO V1. There is no separate phased rollout — the feature is included in the initial product release. Rollout is controlled by feature flag `dws_prefill_enabled` for rapid kill-switch.

Gate	Criteria	Owner
Pre-launch	Zero P0/P1 bugs in golden test cases GTC-001 through GTC-006	Engineering
Pre-launch	DWS-SI-08 domain block-list reviewed and approved by PM + Legal	PM
Pre-launch	Business-name fuzzy match thresholds validated on real dealer names	Engineering + PM
Pre-launch	IM provisioning view updated to show scraped preview column	Engineering
Pre-launch	“Use a different website” UI component built and tested	Engineering + UX
Post-launch monitoring	Scrape success rate $\geq 50\%$; timeout rate $\leq 10\%$; Dealer Admin edit rate $< 25\%$	PM
Rollback trigger	P0: any scrape that writes to a dealer website; any prefilled value saved without Dealer Admin confirmation	Engineering on-call

Feature flag: `dws_prefill_enabled` — set to `false` disables all scraping immediately. No data migration needed for rollback.

Appendix A: ISD Versioning

Version	Date	Author	Change Summary
1.0	Aug 27, 2026	Prem Bihari (pbihari@tekion.com)	Initial ISD draft via paper-mcp isd_init (standard)

Appendix B: PM-002 Scoring Rubric — Standard Mode (Reference)

ISD Standard uses the same 13-section base rubric as the full template (100 base points) plus an 8-point v4.0 autonomy bonus for the two included v4.0 sections.

Section	Weight	Full Credit (100%)	Partial Credit (60%)	No Credit (0%)
§1 Metadata	5pts	All fields complete	Missing optional fields	Missing required fields
§2 Intent Definition	10pts	All criteria defined	Success criteria only	No meaningful definition
§3 NL Triggers	8pts	>= 5 triggers, mix, disambiguation	3-4 triggers	< 3 triggers
§4 Parameters	7pts	All params with source and fallback	Sources defined	Parameter list only
§5.1 Sub-Intents	15pts	IMS Recon granularity; all columns	Happy path only	< 5 sub-intents
§6 Capabilities/Skills	8pts	Business- language caps; agent clear	Caps listed, eng- language	Missing

§7 Business Rules	10pts	ECI contracts; policies identified; §7.5 present	ECI thresholds only	No safety model
§8 Rendering	5pts	Mode declared for all outputs	Primary output only	No declaration
§9 Conversations	8pts	Happy path + edge cases + catalog	Happy path only	No scenarios
§10 Metrics	8pts	KPIs + baselines + attribution	KPIs without baselines	No metrics
§11 Follow-up patterns	5pts	>= 2 patterns	1 pattern	None
§12 Technical Approach	6pts	Constraints + integrations + NFRs	Constraints only	Absent
§13 Rollout Plan	5pts	Phase gates + flags + rollback	Phase list	Absent

Autonomous Agent Bonus: §1.2 Autonomy Declaration (5pts) and §3.4 Event Triggers (3pts). Gate: Score >= 70 required for isd_export. Grand total: 108 max.

7.3 Escalation Rules

Condition	Escalation Path	Notification Target
Scrape consistently fails for a dealership across 3+ sessions	Log to TAO ops dashboard for review; do not alert Dealer Admin	TAO ops team (internal only)
Dealer Admin manually overrides all prefilled fields in 3+ consecutive sessions	Flag pattern to PM team for review	TAO product analytics

7.4 Compliance & Legal Constraints

Web scraping a publicly accessible website generally does not raise legal issues when done for reading the business's own published public information. However:

- TAO must respect `robots.txt` directives on the dealer's website.
- Only the dealer's own website URL may be scraped — no third-party lookups for MVP.
- No personal data is scraped; all scraped fields are business-level public details.
- Legal review recommended before GA if scope expands beyond MVP fields.

7.5 Data Access Constraints

Required per ORBIT S26 I227. Specify RBAC rules, PII handling, and cross-dealer isolation.

Constraint Type	Rule	Applies To	Notes
RBAC	Dealer Admin can view and confirm prefilled details and provide alternate URL. IM can view read-only scraped preview and cannot confirm or save.	DWS-SI-00, DWS-SI-04, DWS-SI-06	Module Champions and PSMs do not interact with either surface
PII Handling	Scraped fields are business-level public details, not personal data.	All sub-intents	If future fields include individual contact names, PII policy must be revisited
Cross-Dealer Isolation	Scrape cache is keyed by (dealership_id, session_id) — no cross-dealer data leakage	DWS-SI-01 through DWS-SI-06	
Data Residency	Scraped data cached and stored in US region only for MVP	DWS-SI-02, DWS-SI-03	Align with TAO's existing data residency policy

7.6 Known Risks

Risk ID	Description	Likelihood	Impact	Mitigation	Owner
R-001	Dealer website is not publicly accessible, behind auth, or has no contact page	High	Low	Silent fallback; no error shown to Dealer Admin	Engineering
R-002	Scraped values are wrong or outdated	Medium	Medium	Confidence gate; Dealer Admin always confirms	PM + UX
R-003	Dealer website blocks scraping via robots.txt	Medium	Low	robots.txt check before scraping; silent skip	Engineering
R-004	Scrape is too slow (>5s) and blocks screen load	Low	High	Hard 5s timeout; async execution	Engineering
R-005	Dealer has no website at all	High for small dealers	None	No-website is a first-class fallback	PM + Engineering
R-006	Dealer Admin provides an intentionally wrong URL	Low to Medium	High	Domain block-list + business-name fuzzy match	Engineering + PM

R-007	Dealer Admin provides group HQ site instead of rooftop site	Medium	Medium	Borderline match prompt for 0.70–0.85	Engineering
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7.10 Financial Controls

Required for financial process intents. Define tolerance bands, 3-way match rules, GL coding, reconciliation gates, segregation of duties, and audit trail requirements. Skip if not applicable.

N/A — this intent does not handle financial transactions. Dealer website scraping is a data pre-fill feature; no monetary values, GL coding, or financial reconciliation are involved.

§8. Rendering Declaration

8.1 Rendering Mode

Sub-Intent / Response Type	Preferred Mode	Fallback	Required Components
Prefilled fields on Dealer Details Review screen	agui (inline form with pre-populated fields, editable)	basic	Prefill indicator label; confidence badge; editable inputs; confirm/submit button
Scrape failure message	basic (inline advisory text, non-blocking)	text	“We couldn't find your details automatically — please fill in below” label
Manual re-scrape trigger (/prefill-dealer-details)	basic (status feedback)	text	Inline spinner or progress indicator; update fields when complete

8.2 Orchestration Handoffs

When this intent hands control to another agent mid-flow, declare the handoff contract here.

Handoff Trigger	Target Agent / Intent	Context Passed	Return Expected	Handoff Type	Notes
Dealer Admin confirms dealer details	TAO Configure Phase — next step in journey	Confirmed dealer detail fields, dealership_id, session_id	Yes — proceed to next TAO step	Blocking	Standard TAO step-completion handoff

§9. Conversation Design & Test Scenarios

9.1 Happy Path Scenarios

#	Trigger	Agent Does	Screen Result
1	IM opens provisioning queue; dealer has website URL in contract	DWS-SI-00 → DWS-SI-00b → builds comparison map	IM sees comparison preview inline with matches and diffs
2	Dealer Admin opens screen; scrape result cached from IM pre-scrape	DWS-SI-01a → DWS-SI-03 → DWS-SI-04	Dealer Admin sees prefilled fields, diff rows, and confirms final values
3	Dealer Admin opens screen; small dealer, no website URL	DWS-SI-01a → DWS-SI-01b → DWS-SI-05	All fields populated from contract; no comparison rows; no mention of websites
4	Website scrape finds same values as contract on all fields	DWS-SI-03 classifies all fields as Match	Quiet “✓ Matches your website” indicators; no comparison rows

5	Dealer Admin provides alternate URL with borderline fuzzy match	DWS-SI-08 shows confirmation prompt	Dealer Admin confirms whether the found website belongs to their dealership
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9.2 Edge Case Scenarios

#	Trigger	Condition	Agent Does	Screen Result
1	Dealer Admin opens screen	No website	DWS-SI-01b finds no URL → contract-only view	All fields pre-populated from contract; no comparison rows
2	Dealer Admin opens screen	Website returns 403 / behind auth	Scrape fails → DWS-SI-05	Same as no-website case
3	Website scrape runs	Phone confidence 0.62	DWS-SI-03 suppresses phone	Phone comparison row does not appear
4	Scrape times out	5s timeout	DWS-SI-05	Screen loads immediately from contract data; no spinner blocking Dealer Admin
5	Dealer Admin provides alternate URL	Domain on block-list	DWS-SI-08 rejects immediately	Shows “This doesn't look like a dealership website.”
6	Dealer Admin provides wrong dealership URL	Business name mismatch	DWS-SI-08 rejects post-scrape	Shows “The website you provided doesn't appear to be for [Dealership Name].”
7	Dealer Admin opens screen	Contract sync has not	DWS-SI-01a cannot load	Screen shows loading state;

		completed	contract data	Dealer Admin cannot proceed
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9.3 Edge Case Catalog

ID	Condition	Expected Behavior	Severity	Test Type
EC-001	Contract sync not complete	Show loading state; block Dealer Admin until sync completes	Blocking	Dependency
EC-002	dealer_website_url is null	Skip scraping layer; render contract-only view	Non-blocking	Negative
EC-003	Dealer website behind auth or scrape fails	Scraping skipped; contract-only view shown	Non-blocking	Boundary
EC-004	robots.txt disallows scraping	Skip scrape silently	Safety Gate	Security
EC-005	All scraped fields match contract data	Suppress comparison rows; show quiet match indicators	Non-blocking	Happy path variant
EC-006	User-provided URL on domain block-list	Reject immediately	Blocking	Security
EC-007	User-provided URL similarity < 0.70	Reject with mismatch message	Blocking	Data quality

EC-008	Borderline match (0.70–0.85)	Show confirmation prompt	Warning	Data quality
EC-009	Dealer Admin submits with required fields empty	Block submit; required field validation error	Blocking	Validation
EC-010	Dealer Admin re-triggers scrape within 5 minutes	Reject with “Please wait before retrying”	Non-blocking	Rate limit

9.4 Golden Test Cases

Minimum 5 golden test cases per intent (Agentic Testing Principle 3).

Sourced from real production scenarios.

GTC#	Input	Expected Output	Quality Dimensions	Pass Criteria
GTC-001	Dealer with website; address differs from contract and website URL is found	Address diff row and “New from website” URL row shown; Dealer Admin accepts desired values	Accuracy; Completeness	Correct diffs surfaced; no false-positive diff rows
GTC-002	Small dealer; dealer_website_url=null	All fields pre-populated from contract; no comparison rows	No-website baseline	Screen fully prefilled; zero website UI
GTC-003	Website response time 6s	Screen loads from contract data immediately; scrape times out silently	Timeout enforcement	Screen visible in < 500ms; no error

GTC-004	Dealer Admin provides blocked domain	DWS-SI-08 rejects immediately; comparison view unchanged	Domain validation	Immediate rejection; contract prefill unaffected
GTC-005	Dealer Admin provides competitor URL	DWS-SI-08 rejects post-scrape; contract prefill unaffected	Business-name fuzzy match	Rejection message shown
GTC-006	trigger_surface="im_provisioning"; address diff found	IM dealer list row shows contract values + address diff highlighted	IM preview	IM sees diff inline; no IM action/save possible

9.5 Quality Dimensions

Explicit quality dimensions with thresholds for every LLM-generated output (Principle 2).

Dimension	Description	Target Threshold	Regression Threshold	Eval Method
Accuracy	Factual correctness of responses	>= 95%	< 90% blocks	Deterministic + LLM-as-judge
Completeness	All required information included	>= 90%	< 85% blocks	Schema validation
Tone	Professional, contextually appropriate	>= 4/5	< 3/5 flags	LLM-as-judge
Safety	No harmful, private, or hallucinated content	100%	< 100% blocks	Deterministic + red-team

9.6 Failure Mode Coverage

Failure Type	Applicable?	Specific Risk	Mitigation	Test Strategy
Logic	Y	Field mapping may misidentify fields on non-standard dealer websites	Confidence scoring + field-type validation	Unit tests with diverse HTML fixtures
Integration	Y	Website unreachable, slow, redirecting, or unexpected content type	Timeout, robots.txt check, MIME type guard	Pipeline tests with mock HTTP responses
Behavioral	N	No LLM reasoning in happy path	N/A	N/A
Quality	Y	Scraped values may be outdated or incorrect	Confidence threshold; Dealer Admin always confirms	Golden set with known-correct vs. outdated websites
Drift	Y	Dealer website HTML structure changes over time	Monitor scrape success rates	Weekly regression on dealer URL golden set
Context	N	Intent scoped to Dealer Details Review screen	N/A	N/A
Safety	Y	Scrape must not write to, submit forms on, or authenticate with dealer website	Read-only HTTP GET; no cookies/session storage	Red-team verification that no POST/PUT/auth calls are made

Coordination	N	Intent operates independently	N/A	N/A
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§10. Success Metrics & Evaluation

10.1 Business KPIs

KPI	Current Baseline	Target	Measurement Method	Timeline
Blank-field rate at screen open	~100% (all fields blank today)	0%	TAO analytics — blank screen loads / total loads	At TAO V1 launch
Dealer Details screen completion time	TBD — measure at TAO V1 launch	Reduce by >= 40% vs. baseline	TAO analytics — screen open to submit	30 days post-launch
Dealer Admin field correction rate	N/A	< 15% of contract-prefilled fields manually corrected	DA_edit events / total contract fields shown	30 days post-launch
Website diff acceptance rate	N/A	Target >= 50%	Accepted website diffs / total diff rows shown	30 days post-launch
Dealer Details form validation error rate	TBD — measure at TAO V1 launch	Reduce by >= 60% vs. baseline	TAO analytics — validation errors on submit	30 days post-launch

10.2 Agent Evaluation Criteria

Metric	Definition	Target	Regression Threshold	Cadence
Intent match rate	% correctly classified	> 95%	< 90% triggers review	Weekly

Golden set pass rate	% golden scenarios passing	> 98%	< 95% blocks release	Per deploy
Quality dimension scores	Per-dimension eval scores	Per §9.5 targets	Per §9.5 regression	Per deploy
Invariant violation rate	% operations violating invariants	0%	> 0% blocks release	Continuous

10.3 Invariants (Continuous Checks)

Properties that must ALWAYS hold — checked on every operation, not periodically (Principle 5).

Invariant	Description	Check Frequency	Enforcement
Layer 1 must precede Layer 2	Dealer Admin screen must never render without Layer 1 contract data present	Every screen load	Hard block
Screen is never blank	Contract values fill every field; Layer 2 cannot make a field emptier than Layer 1 left it	Every render	Hard block
Read-only scrape	Scraper never submits forms, writes cookies, or authenticates	Every scrape	Hard block
Dealer Admin confirmation required before save	No value persisted without explicit Dealer Admin submit action	Every save	Hard block

Dealer isolation	Scrape/comparison results for one dealership never shown to another	Every render	Hard block
URL validation before user-provided scrape	DWS-SI-02 must never run on dealer_provided_url unless DWS-SI-08 passed it	Every relevant scrape	Hard block
IM cannot save dealer details	IM surface is read-only; write endpoint rejects IM provisioning saves	Every write	Hard block
Data integrity	No orphaned references, append-only IDs	Every write	Hard block
Privacy	No private content in external outputs	Every export	Hard block

10.4 Related Intents & Dependencies

Upstream Dependencies

Intent ID	Intent Name	Dependency Type	Notes
TAO-CONTRACT-001 (implicit)	Contract Signature & Dealer Intake	Data	dealer_website_url must be captured at contract signature
TAO-COA-001	COA / Financial Statement Upload (Step 7)	Workflow	Same Configure Phase; no direct dependency

Downstream Effects

Intent ID	Intent Name	Effect Type	Notes

TAO Configure Phase — next step	Dealer Details confirmed → Configure Phase proceeds	Workflow	DWS-001 completion unblocks the next step
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§11. Follow-Up Patterns

Pattern	Trigger	Agent Response
Dealer Admin edits a prefilled field	Dealer Admin changes a prefilled value in the form	Accept the edit silently; edited value replaces prefilled value — Dealer Admin is always the source of truth
Dealer Admin requests re-scrape	Dealer Admin triggers /prefill-dealer-details	Check debounce window; if allowed, re-run DWS-SI-02 through DWS-SI-04; if within debounce, show “Please wait before retrying”
Prefill partially succeeds	DWS-SI-03 returns a partial field map	Prefill found fields; leave unfound fields blank with standard placeholder text; do not show partial-failure message

§12. Technical Approach

Standard feature mode — engineering-review format. AI capability mapping not required.

Architecture — Two-Layer Model

This feature has two independent layers that compose on the Dealer Details Review screen.

Layer 1 — Internal Data Sync (baseline, always runs)

Data flow: `Salesforce → TOC → TAO`

1. **TOC Sync Service** — At IM provisioning stage, reads the dealer record from TOC. TAO does not connect to Salesforce directly — TOC is the single API call.
 2. **TAO Dealer Record** — Stores the TOC-sourced dealer record. Dealer Admin screen reads from here. Always populated before the Dealer Admin journey begins.
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Layer 2 — Website Scraping + Comparison (enrichment, additive)

3. **Scrape Trigger** — Resolves the dealer website URL from the TAO dealer record. If no URL, Layer 2 is skipped entirely.
 4. **Web Scraper Worker** — HTTP GET to target URL (homepage + contact/about page). Enforces 5s timeout, robots.txt, MIME type guard, and read-only GET.
 5. **Field Extractor** — Parses raw HTML and extracts dealership name, DBA, address, phones, email, operating hours, and logo URL with confidence scores.
 6. **Field Comparator** — Compares extracted values against contract values and classifies fields as match, different, new, or missing.
 7. **Comparison View Renderer** — Injects comparison map into the Dealer Details Review screen.
 8. **Session Cache** — Caches comparison result keyed by (dealership_id, session_id).
 9. **TAO Dealer Record Write** — On Dealer Admin submit, confirmed final values are persisted.
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URL Validation (for dealer-admin-provided alternate URLs)

10. **URL Validator** — Domain block-list check; scrape and fuzzy-match extracted business name against contract dealership name. Below 0.70 rejects, 0.70–0.85 prompts, and ≥ 0.85 accepts silently.
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Non-Functional Requirements

- **Screen load latency:** Dealer Admin screen must render from Layer 1 data in $< 500\text{ms}$.
 - **Layer 1 is blocking:** Dealer Admin cannot open the Dealer Details Review screen until DWS-SI-00 is complete.
 - **Layer 2 is always additive:** Any Layer 2 failure falls back silently to Layer 1 data only.
 - **Rate limiting:** Max 1 scrape per (dealership_id, session); max 1 user-triggered re-scrape per 5 minutes per dealership.
 - **Security:** Read-only HTTP GET only. No auth. No cookies. No form submissions.
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§13. Rollout Plan

Standard feature mode — TAO V1 launch.

Rollout Target: TAO V1

This feature ships with TAO V1. There is no separate phased rollout — the feature is included in the initial product release. Rollout is controlled by feature flag `dws_prefill_enabled` for rapid kill-switch.

Gate	Criteria	Owner
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Pre-launch	Business-name fuzzy match thresholds validated on real dealer names	Engineering + PM
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Autonomous Agent Bonus: §1.2 Autonomy Declaration (5pts) and §3.4 Event Triggers (3pts). Gate: Score >= 70 required for isd_export. Grand total: 108 max.

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Condition	Escalation Path	Notification Target
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§8. Rendering Declaration

8.1 Rendering Mode

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Manual re-scrape trigger (/prefill-dealer-details)	basic (status feedback)	text	Inline spinner or progress indicator; update fields when complete

8.2 Orchestration Handoffs

When this intent hands control to another agent mid-flow, declare the handoff contract here.

Handoff Trigger	Target Agent / Intent	Context Passed	Return Expected	Handoff Type	Notes
Dealer Admin confirms dealer details	TAO Configure Phase — next step in journey	Confirmed dealer detail fields, dealership_id, session_id	Yes — proceed to next TAO step	Blocking	Standard TAO step-completion handoff

§9. Conversation Design & Test Scenarios

9.1 Happy Path Scenarios

#	Trigger	Agent Does	Screen Result
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3	Dealer Admin opens screen; small dealer, no website URL	DWS-SI-01a → DWS-SI-01b → DWS-SI-05	All fields populated from contract; no comparison rows; no mention of websites
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EC-004	robots.txt disallows scraping	Skip scrape silently	Safety Gate	Security
EC-005	All scraped fields match contract data	Suppress comparison rows; show quiet match indicators	Non-blocking	Happy path variant
EC-006	User-provided URL on domain block-list	Reject immediately	Blocking	Security
EC-007	User-provided URL similarity < 0.70	Reject with mismatch message	Blocking	Data quality

EC-008	Borderline match (0.70–0.85)	Show confirmation prompt	Warning	Data quality
EC-009	Dealer Admin submits with required fields empty	Block submit; required field validation error	Blocking	Validation
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Minimum 5 golden test cases per intent (Agentic Testing Principle 3).
Sourced from real production scenarios.

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GTC-001	Dealer with website; address differs from contract and website URL is found	Address diff row and “New from website” URL row shown; Dealer Admin accepts desired values	Accuracy; Completeness	Correct diffs surfaced; no false-positive diff rows
GTC-002	Small dealer; dealer_website_url=null	All fields pre-populated from contract; no comparison rows	No-website baseline	Screen fully prefilled; zero website UI
GTC-003	Website response time 6s	Screen loads from contract data immediately; scrape times out silently	Timeout enforcement	Screen visible in < 500ms; no error

GTC-004	Dealer Admin provides blocked domain	DWS-SI-08 rejects immediately; comparison view unchanged	Domain validation	Immediate rejection; contract prefill unaffected
GTC-005	Dealer Admin provides competitor URL	DWS-SI-08 rejects post-scrape; contract prefill unaffected	Business-name fuzzy match	Rejection message shown
GTC-006	trigger_surface="im_provisioning"; address diff found	IM dealer list row shows contract values + address diff highlighted	IM preview	IM sees diff inline; no IM action/save possible

9.5 Quality Dimensions

Explicit quality dimensions with thresholds for every LLM-generated output (Principle 2).

Dimension	Description	Target Threshold	Regression Threshold	Eval Method
Accuracy	Factual correctness of responses	>= 95%	< 90% blocks	Deterministic + LLM-as-judge
Completeness	All required information included	>= 90%	< 85% blocks	Schema validation
Tone	Professional, contextually appropriate	>= 4/5	< 3/5 flags	LLM-as-judge
Safety	No harmful, private, or hallucinated content	100%	< 100% blocks	Deterministic + red-team

9.6 Failure Mode Coverage

Failure Type	Applicable?	Specific Risk	Mitigation	Test Strategy
Logic	Y	Field mapping may misidentify fields on non-standard dealer websites	Confidence scoring + field-type validation	Unit tests with diverse HTML fixtures
Integration	Y	Website unreachable, slow, redirecting, or unexpected content type	Timeout, robots.txt check, MIME type guard	Pipeline tests with mock HTTP responses
Behavioral	N	No LLM reasoning in happy path	N/A	N/A
Quality	Y	Scraped values may be outdated or incorrect	Confidence threshold; Dealer Admin always confirms	Golden set with known-correct vs. outdated websites
Drift	Y	Dealer website HTML structure changes over time	Monitor scrape success rates	Weekly regression on dealer URL golden set
Context	N	Intent scoped to Dealer Details Review screen	N/A	N/A
Safety	Y	Scrape must not write to, submit forms on, or authenticate with dealer website	Read-only HTTP GET; no cookies/session storage	Red-team verification that no POST/PUT/auth calls are made

Coordination	N	Intent operates independently	N/A	N/A
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§10. Success Metrics & Evaluation

10.1 Business KPIs

KPI	Current Baseline	Target	Measurement Method	Timeline
Blank-field rate at screen open	~100% (all fields blank today)	0%	TAO analytics — blank screen loads / total loads	At TAO V1 launch
Dealer Details screen completion time	TBD — measure at TAO V1 launch	Reduce by >= 40% vs. baseline	TAO analytics — screen open to submit	30 days post-launch
Dealer Admin field correction rate	N/A	< 15% of contract-prefilled fields manually corrected	DA_edit events / total contract fields shown	30 days post-launch
Website diff acceptance rate	N/A	Target >= 50%	Accepted website diffs / total diff rows shown	30 days post-launch
Dealer Details form validation error rate	TBD — measure at TAO V1 launch	Reduce by >= 60% vs. baseline	TAO analytics — validation errors on submit	30 days post-launch

10.2 Agent Evaluation Criteria

Metric	Definition	Target	Regression Threshold	Cadence
Intent match rate	% correctly classified	> 95%	< 90% triggers review	Weekly

Golden set pass rate	% golden scenarios passing	> 98%	< 95% blocks release	Per deploy
Quality dimension scores	Per-dimension eval scores	Per §9.5 targets	Per §9.5 regression	Per deploy
Invariant violation rate	% operations violating invariants	0%	> 0% blocks release	Continuous

10.3 Invariants (Continuous Checks)

Properties that must ALWAYS hold — checked on every operation, not periodically (Principle 5).

Invariant	Description	Check Frequency	Enforcement
Layer 1 must precede Layer 2	Dealer Admin screen must never render without Layer 1 contract data present	Every screen load	Hard block
Screen is never blank	Contract values fill every field; Layer 2 cannot make a field emptier than Layer 1 left it	Every render	Hard block
Read-only scrape	Scraper never submits forms, writes cookies, or authenticates	Every scrape	Hard block
Dealer Admin confirmation required before save	No value persisted without explicit Dealer Admin submit action	Every save	Hard block

Dealer isolation	Scrape/comparison results for one dealership never shown to another	Every render	Hard block
URL validation before user-provided scrape	DWS-SI-02 must never run on dealer_provided_url unless DWS-SI-08 passed it	Every relevant scrape	Hard block
IM cannot save dealer details	IM surface is read-only; write endpoint rejects IM provisioning saves	Every write	Hard block
Data integrity	No orphaned references, append-only IDs	Every write	Hard block
Privacy	No private content in external outputs	Every export	Hard block

10.4 Related Intents & Dependencies

Upstream Dependencies

Intent ID	Intent Name	Dependency Type	Notes
TAO-CONTRACT-001 (implicit)	Contract Signature & Dealer Intake	Data	dealer_website_url must be captured at contract signature
TAO-COA-001	COA / Financial Statement Upload (Step 7)	Workflow	Same Configure Phase; no direct dependency

Downstream Effects

Intent ID	Intent Name	Effect Type	Notes

TAO Configure Phase — next step	Dealer Details confirmed → Configure Phase proceeds	Workflow	DWS-001 completion unblocks the next step
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§11. Follow-Up Patterns

Pattern	Trigger	Agent Response
Dealer Admin edits a prefilled field	Dealer Admin changes a prefilled value in the form	Accept the edit silently; edited value replaces prefilled value — Dealer Admin is always the source of truth
Dealer Admin requests re-scrape	Dealer Admin triggers /prefill-dealer-details	Check debounce window; if allowed, re-run DWS-SI-02 through DWS-SI-04; if within debounce, show “Please wait before retrying”
Prefill partially succeeds	DWS-SI-03 returns a partial field map	Prefill found fields; leave unfound fields blank with standard placeholder text; do not show partial-failure message

§12. Technical Approach

Standard feature mode — engineering-review format. AI capability mapping not required.

Architecture — Two-Layer Model

This feature has two independent layers that compose on the Dealer Details Review screen.

Layer 1 — Internal Data Sync (baseline, always runs)

Data flow: Salesforce → TOC → TAO

1. **TOC Sync Service** — At IM provisioning stage, reads the dealer record from TOC. TAO does not connect to Salesforce directly — TOC is the single API call.
 2. **TAO Dealer Record** — Stores the TOC-sourced dealer record. Dealer Admin screen reads from here. Always populated before the Dealer Admin journey begins.
-

Layer 2 — Website Scraping + Comparison (enrichment, additive)

3. **Scrape Trigger** — Resolves the dealer website URL from the TAO dealer record. If no URL, Layer 2 is skipped entirely.
 4. **Web Scraper Worker** — HTTP GET to target URL (homepage + contact/about page). Enforces 5s timeout, robots.txt, MIME type guard, and read-only GET.
 5. **Field Extractor** — Parses raw HTML and extracts dealership name, DBA, address, phones, email, operating hours, and logo URL with confidence scores.
 6. **Field Comparator** — Compares extracted values against contract values and classifies fields as match, different, new, or missing.
 7. **Comparison View Renderer** — Injects comparison map into the Dealer Details Review screen.
 8. **Session Cache** — Caches comparison result keyed by (dealership_id, session_id).
 9. **TAO Dealer Record Write** — On Dealer Admin submit, confirmed final values are persisted.
-

URL Validation (for dealer-admin-provided alternate URLs)

10. **URL Validator** — Domain block-list check; scrape and fuzzy-match extracted business name against contract dealership name. Below 0.70 rejects, 0.70–0.85 prompts, and ≥ 0.85 accepts silently.
-

Non-Functional Requirements

- **Screen load latency:** Dealer Admin screen must render from Layer 1 data in $< 500\text{ms}$.
 - **Layer 1 is blocking:** Dealer Admin cannot open the Dealer Details Review screen until DWS-SI-00 is complete.
 - **Layer 2 is always additive:** Any Layer 2 failure falls back silently to Layer 1 data only.
 - **Rate limiting:** Max 1 scrape per (dealership_id, session); max 1 user-triggered re-scrape per 5 minutes per dealership.
 - **Security:** Read-only HTTP GET only. No auth. No cookies. No form submissions.
-

§13. Rollout Plan

Standard feature mode — TAO V1 launch.

Rollout Target: TAO V1

This feature ships with TAO V1. There is no separate phased rollout — the feature is included in the initial product release. Rollout is controlled by feature flag `dws_prefill_enabled` for rapid kill-switch.

Gate	Criteria	Owner
Pre-launch	Zero P0/P1 bugs in golden test cases GTC-001 through GTC-006	Engineering
Pre-launch	DWS-SI-08 domain block-list reviewed and approved by PM + Legal	PM
Pre-launch	Business-name fuzzy match thresholds validated on real dealer names	Engineering + PM
Pre-launch	IM provisioning view updated to show scraped preview column	Engineering
Pre-launch	“Use a different website” UI component built and tested	Engineering + UX
Post-launch monitoring	Scrape success rate $\geq 50\%$; timeout rate $\leq 10\%$; Dealer Admin edit rate $< 25\%$	PM
Rollback trigger	P0: any scrape that writes to a dealer website; any prefilled value saved without Dealer Admin confirmation	Engineering on-call

Feature flag: `dws_prefill_enabled` — set to `false` disables all scraping immediately. No data migration needed for rollback.

Appendix A: ISD Versioning

Version	Date	Author	Change Summary
1.0	Aug 27, 2026	Prem Bihari (pbihari@tekion.com)	Initial ISD draft via paper-mcp isd_init (standard)

Appendix B: PM-002 Scoring Rubric — Standard Mode (Reference)

ISD Standard uses the same 13-section base rubric as the full template (100 base points) plus an 8-point v4.0 autonomy bonus for the two included v4.0 sections.

Section	Weight	Full Credit (100%)	Partial Credit (60%)	No Credit (0%)
§1 Metadata	5pts	All fields complete	Missing optional fields	Missing required fields
§2 Intent Definition	10pts	All criteria defined	Success criteria only	No meaningful definition
§3 NL Triggers	8pts	>= 5 triggers, mix, disambiguation	3-4 triggers	< 3 triggers
§4 Parameters	7pts	All params with source and fallback	Sources defined	Parameter list only
§5.1 Sub-Intents	15pts	IMS Recon granularity; all columns	Happy path only	< 5 sub-intents
§6 Capabilities/Skills	8pts	Business-language caps; agent clear	Caps listed, eng-language	Missing

§7 Business Rules	10pts	ECI contracts; policies identified; §7.5 present	ECI thresholds only	No safety model
§8 Rendering	5pts	Mode declared for all outputs	Primary output only	No declaration
§9 Conversations	8pts	Happy path + edge cases + catalog	Happy path only	No scenarios
§10 Metrics	8pts	KPIs + baselines + attribution	KPIs without baselines	No metrics
§11 Follow-up patterns	5pts	>= 2 patterns	1 pattern	None
§12 Technical Approach	6pts	Constraints + integrations + NFRs	Constraints only	Absent
§13 Rollout Plan	5pts	Phase gates + flags + rollback	Phase list	Absent

Autonomous Agent Bonus: §1.2 Autonomy Declaration (5pts) and §3.4 Event Triggers (3pts).
Gate: Score >= 70 required for isd_export. Grand total: 108 max.

ORBIT ISD | Intent ID: TAO-DWS-001 | PM-002 Score: 94/100 | Gate: PASS_GA | Status: Draft | Owner: pbihari@tekion.com | Eng Lead: Nitish Kumar | Review Date: 1 Sep 2026

§1. Document Metadata

Field	Value
Intent ID	TAO-DWS-001
Intent Name	Dealer Website Scraping — Pre-fill Dealer Details Review

Product Area	Dealer Onboarding / TAO
Module / Sub-Area	TAO Configure Phase — Dealer Details Review Screen
Document Owner (PM)	pbihari@tekion.com
Engineering Lead	Nitish Kumar
Status	Draft
Version	1.0
Last Updated	2026-08-27
Review Date	1st September 2026
Source Reference	TAO_User_Journey.md — Dealer Details Review screen
Linked Persona(s)	Dealer Admin
ORBIT Template Version	ISD-v4.5-standard

1.2 Autonomy Declaration

Field	Value
Execution Model	hybrid — event-driven at Dealer Admin screen load AND proactive at IM provisioning stage
Autonomy Tier (ORBIT S04)	A1 (Suggest-Only) — scraper proposes prefilled values; human must confirm at every surface
Target Autonomy Tier (GA)	A1 (suggest-only) — confirmation always required; no auto-save of scraped data
Trigger Classification	event-triggered (Dealer Admin screen load) + proactive (IM provisioning stage)

Context Persistence	session-scoped for Dealer Admin path; provisioning-record-scoped for IM path
Execution Surface	TAO backend service — server-side
Background Execution	Yes — scrape runs in background; result cached; neither Dealer Admin screen nor IM view is blocked

§2. Intent Definition

2.1 Primary Intent and Success Criteria

Field	Value
Primary Intent	Populate the TAO Dealer Details Review screen from two sources — (1) TOC/contract data as the always-present baseline, and (2) website scraping as an enrichment layer — then present a before/after comparison for Dealer Admin confirmation
Success Criteria	(1) Dealer Details Review screen is never blank — TOC/contract data always prefills it. (2) If a website exists, a comparison view shows what the website found vs. what is already on file, field by field. (3) Dealer Admin confirms the final values once; no manual re-typing of data already known to Tekion.
Value	High — eliminates blank-form experience for all dealers regardless of website presence
Golden Intent?	Yes

Automation Potential	Partial — TOC sync and scraping are fully automated; Dealer Admin confirmation is always manual (A1)
Expected Volume	1 TOC-sync invocation per dealer provisioning + 1 scrape per Dealer Admin session

2.2 User Personas

Persona	Role	Primary Use Case
Dealer Admin	Primary dealership-side owner	Reviews, edits, and confirms prefilled dealer details; can provide a corrected website URL
Implementation Manager (IM)	Tekion-side operator	Views scraped dealer details during provisioning to verify info before the Dealer Admin journey begins

2.3 Business Context

The Dealer Details Review screen must never be blank. When a dealer signs a contract with Tekion, their basic details — name, address, phone, website URL — are captured in Salesforce. Salesforce is already integrated with **TOC (Tekion Operating Cloud)** and syncs dealer data into TOC as part of the contracting and provisioning pipeline. **TAO connects to TOC only; TAO does not read from Salesforce directly.**

Data flow: Salesforce → TOC → TAO

Layer 1 — Internal data sync (baseline, always present): At provisioning stage, TAO reads dealer details from TOC (which already holds the Salesforce-synced contract data plus any provisioning-stage updates). When the Dealer Admin opens the Dealer Details Review screen, all fields are already populated. There is no TAO-side merge step — TOC owns the reconciliation.

Layer 2 — Website scraping (enrichment, additive): For dealers with a public website, TAO scrapes it and compares what it finds against the on-file record. The website scrape surfaces differences without discarding what Tekion already knows.

The Dealer Admin experience: When the Dealer Admin opens the screen they see:

- All fields pre-populated from TOC record, labelled "From your records on file"
- Where the website found something different or new: "On file: 123 Main St | Found on website: 123 Main Street, Suite 100 | Use website version?"
- Where the website found the same value → a quiet checkmark
- Where the website found nothing → on-file value shown alone

Small dealers with no website: TOC/contract data fills the screen completely. No comparison rows. This is the ideal baseline — not a degraded experience.

§3. Natural Language Triggers

3.1 Example Triggers

Trigger Phrase	Confidence	Context
Dealer Details Review screen load event (system trigger)	High	Fires automatically on screen load
"Prefill my dealership info"	High	Dealer Admin requests prefill from website
"Use my website to fill in these details"	High	Dealer Admin provides intent
"Look up my dealership"	Medium	Could be website scrape or internal lookup
"Fill in my address from my website"	Medium	Field-specific request

3.2 Disambiguation Notes

This intent is event-triggered (screen load), not conversational in the primary path. Resolve ambiguous lookup requests by: checking if dealer has a website URL on file; if yes, default to

website scrape.

3.3 Slash Commands

Command	Description
/prefill-dealer-details	Manually trigger a website scrape to re-populate the screen

3.4 Event Triggers

Trigger Type	Event Source	Condition	Debounce
IM provisioning stage load	TAO IM portal	Dealer record enters provisioning state AND dealer_website_url is present	Max 1 scrape per dealer provisioning record
Dealer Admin screen-load event	TAO frontend — Dealer Details Review screen	Screen renders for the first time in this session AND dealer_website_url is available	Max 1 auto-scrape per session
Dealer Admin provides alternate URL	Dealer Admin input on screen	Dealer Admin submits a URL AND URL passes validation (DWS-SI-08)	Max 1 per 5 minutes per dealership
Manual re-scrape (/prefill-dealer-details)	Dealer Admin-invoked	Dealer Admin explicitly requests re-scrape	Max 1 per 5 minutes per dealership

§4. Parameters

Parameter	Type	Required	Source	If Missing
dealership_id	string	Required	TAO session context	Block execution
session_id	string	Required	TAO session context	Block execution
toc_dealer_data	object	Required	TOC dealer record (single source; holds Salesforce-synced contract data + provisioning updates)	Block execution — screen cannot render without TOC baseline
dealer_website_url	string (URL)	Optional	TOC dealer record (from Salesforce contract at deal close)	Skip scraping layer; screen renders from toc_dealer_data
dealer_provided_url	string (URL)	Optional	Dealer Admin input ("Try a different website")	N/A — must pass DWS-SI-08 validation
scrape_timeout_ms	number	Optional	TAO backend config (default: 5000ms)	Use default 5000ms
fields_to_compare	string[]	Optional	TAO configuration	Default: dealership_name, address, city, state, zip,

				phone, website_url
trigger_surface	enum	Required	System context	Default to "da_screen"

§5. Execution Flow

5.1 Sub-Intents

Sub-Intent	Description	Trigger Condition	Safety Gate	Phase
DWS-SI-00: Sync TOC dealer data to TAO dealer record	Read dealer details from TOC; write into TAO dealer record. No TAO-side merge — TOC is single source.	Dealer enters provisioning state	Required — blocks Dealer Admin screen load if not complete	MVP
DWS-SI-00b: IM provisioning scrape preview	After TOC sync, proactively scrape dealer website (if URL present) and surface comparison preview in IM dealer list view	DWS-SI-00 completes AND dealer_website_url is present	IM preview is read-only; IM cannot confirm or save	MVP
DWS-SI-01a: Load internal	On Dealer Admin	Dealer Admin Dealer	Required — screen must	MVP

data prefill	screen load, read TOC-sourced dealer record and render as baseline prefill	Details Review screen loads	never be blank; blocks if DWS-SI-00 has not run	
DWS-SI-01b: Resolve dealer website URL	Look up dealer website URL from TOC record; check for IM pre-scrape cache	DWS-SI-01a completes	If no URL → skip to DWS-SI-04 (contract-only view)	MVP
DWS-SI-02: Scrape public website	Fetch and parse dealer website; read-only GET only; 5s timeout	URL resolved OR Dealer Admin-provided URL after DWS-SI-08 passes	Rate limit; robots.txt check; read-only GET only; scrape failure → silent fallback	MVP
DWS-SI-03: Compare scraped data against on-file data	For each field: classify as Match / Different / New / Missing	Scrape completed	Confidence gate: only fields ≥ 0.80 confidence enter comparison	MVP
DWS-SI-04: Present comparison view to Dealer Admin	Render baseline + comparison diff rows +	DWS-SI-03 complete OR no scrape ran	A1 gate — Dealer Admin must submit; no value saved	MVP

	match indicators		without explicit Dealer Admin action	
DWS-SI-05: No-scrape path	Dealer Admin sees contract-prefilled screen with no comparison rows; no error shown	No URL found OR scrape fails/times out	None — contract data already shown; no degradation	MVP
DWS-SI-06: Save confirmed dealer details	Persist Dealer Admin-confirmed final values to TAO dealer record	Dealer Admin submits confirmed details	Required field validation; block if mandatory fields empty	MVP
DWS-SI-07: Dealer Admin provides alternate website URL	Dealer Admin inputs a corrected URL via "Try a different website"	Dealer Admin submits a URL	Route to DWS-SI-08 for validation	MVP
DWS-SI-08: Validate user-provided URL	Domain block-list check; post-scrape business-name fuzzy match ≥ 0.70 against contract	DWS-SI-07 fires	Hard gate: rejection or confirmation prompt shown to Dealer Admin	MVP

	dealership name			
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§6. Capabilities, Skills & Agent Assignment

6.1 Agent Assignment

Field	Value
Agent	TAO Onboarding Agent (T1 Base)
Agent Description	Handles dealer-side self-service onboarding steps within TAO; this intent adds a web scraping sub-capability to the Configure Phase screen load sequence

6.2 Capability Needs

Capability Need	Business Description	Data Returned
TOC dealer data read	Read dealer details from TOC dealer record (holds Salesforce-synced contract data + provisioning updates)	Structured dealer details object
TAO dealer record write (internal sync)	Write TOC-sourced record into TAO dealer record at provisioning	Write confirmation
Public website fetch	Retrieve HTML from dealer's public website (homepage + contact/about page)	Raw HTML
HTML text extraction + field compare	Parse HTML, extract fields, assign confidence scores,	Structured comparison map

	compare against on-file data	(Match/Different/New/Missing)
Comparison view render	Render before/after comparison UI on Dealer Details Review screen	Per-field comparison rows
TAO dealer record write (confirmed)	Persist Dealer Admin-confirmed final values after Dealer Admin submits	Confirmation receipt

6.3 Skill Inventory

Skill Name	Knowledge Description	Reusable Across
dealer-web-scrape	Fetches dealer's public website; extracts business name, address, phone, URL fields	Any TAO step needing dealer info from their public website
dealer-details-mapper	Maps raw scraped text to TAO field schema; handles address normalization, phone formatting, confidence scoring	Future TAO screens showing dealer info

6.4 Data Sources

Data Source	Description	Access Pattern
TOC (Tekion Operating Cloud) dealer record	Single source of dealer truth for TAO. Holds Salesforce-synced contract data plus provisioning-stage updates.	Read via TOC API at provisioning time (DWS-SI-00); TAO does not connect to Salesforce directly

	Salesforce→TOC sync is upstream of TAO.	
TAO dealer record	TAO's internal dealer data store; holds the TOC-sourced record used to prefill the Dealer Admin screen	Read at Dealer Admin screen load (DWS-SI-01a); Written at provisioning sync (DWS-SI-00) and after Dealer Admin confirmation (DWS-SI-06)
Dealer's public website	Publicly accessible HTML — enrichment/comparison layer (Layer 2)	Read-only HTTP GET; respects robots.txt; no auth; homepage + contact/about page

6.5 Content Requirements

Content Type	Audience	Format
Internal-data prefill baseline	Dealer Admin	All fields pre-populated from TOC record; labelled "From your records on file"
Comparison diff row	Dealer Admin	"On file: [value] / Found on your website: [value] / Use website version? [Yes] [Keep on file]"
Match indicator	Dealer Admin	Quiet green checkmark: "✓ Matches your website"
IM comparison preview	Implementation Manager	Read-only inline row in dealer list: contract value vs. website value per field

URL validation rejection	Dealer Admin	"The website you provided doesn't appear to be for [Dealership Name]. Please check the URL and try again."
Borderline match confirmation	Dealer Admin	"We found [scraped name] on this website — is this your dealership? [Yes, use this website] [No, try a different URL]"

§7. Business Rules & Safety

7.1 ECI Contract — Action Safety Levels

Sub-Intent	Risk Level	Requires Confirmation	Notes
DWS-SI-00: TOC sync to TAO record	write (internal)	false	IM-triggered; deterministic
DWS-SI-00b: IM provisioning scrape preview	read-only	false	IM sees result; cannot confirm or save
DWS-SI-01a: Load internal data prefill	read-only	false	Required baseline; never skipped
DWS-SI-02: Scrape public website	read-only	false	Read-only GET only
DWS-SI-03: Compare	read-only	false	Fields < 0.80 confidence suppressed

scraped vs. on-file data			
DWS-SI-04: Present comparison view	read-only (display)	true — Dealer Admin must submit to save	A1 gate
DWS-SI-05: No-scrape path	read-only (display)	true — Dealer Admin must still submit	Contract data shown; Dealer Admin confirms
DWS-SI-06: Save confirmed details	write	true — Dealer Admin submit is the trigger	Never auto-saved
DWS-SI-07: Dealer Admin provides alternate URL	read-only (input)	false	Routes to DWS-SI-08
DWS-SI-08: Validate user-provided URL	read-only (validation)	true — result shown to Dealer Admin	Hard gate

7.2 Business Rules — Policy Lake

Policy ID	Rule	Scope
POL-TAO-DWS-001-000	Screen must never be blank: TOC data must be synced to TAO dealer record before Dealer Admin opens screen	DWS-SI-00, DWS-SI-01a
POL-TAO-DWS-001-000a	Data flow is Salesforce → TOC → TAO. TAO reads from TOC only. TOC owns reconciliation of	DWS-SI-00

	Salesforce + provisioning data.	
POL-TAO-DWS-001-001	Scraping is read-only: HTTP GET only — never POST, submit forms, authenticate, or store cookies	All scrape executions
POL-TAO-DWS-001-002	Dealer Admin must explicitly confirm final values before save — no auto-save	DWS-SI-04, DWS-SI-05, DWS-SI-06
POL-TAO-DWS-001-003	No-website is not a degraded state: Dealer Admin sees fully prefilled form from TOC/contract data with no website-related UI	DWS-SI-05
POL-TAO-DWS-001-004	Scrape failure is not a degraded state: on failure, Dealer Admin sees the prefilled form with no comparison rows — no error	DWS-SI-05
POL-TAO-DWS-001-005	Only fields with confidence ≥ 0.80 shown as prefilled without warning; fields below 0.80 display warning badge	DWS-SI-03, DWS-SI-04
POL-TAO-DWS-001-006	Scraping must respect robots.txt: if disallowed, skip scrape silently	DWS-SI-02
POL-TAO-DWS-001-007	User-provided URLs must pass domain	DWS-SI-08

	block-list check before scraping (blocks Google, Facebook, Yelp, etc.)	
POL-TAO-DWS-001-008	User-provided URL scrape must fuzzy-match dealership name ≥ 0.70 ; below threshold → reject with message	DWS-SI-08
POL-TAO-DWS-001-009	IM pre-scraped data is informational only — IM cannot confirm or save dealer details	DWS-SI-00

7.3 Escalation Rules

Condition	Escalation Path	Notification Target
Scrape consistently fails for a dealership across 3+ sessions	Log to TAO ops dashboard	TAO ops team (internal only)
Dealer Admin manually overrides all prefilled fields in 3+ consecutive sessions	Flag to PM for data quality review	TAO product analytics

7.4 Compliance & Legal

- TAO must respect `robots.txt` on the dealer's website.
- Only the dealer's own website URL (from contract) may be scraped — no third-party lookups for MVP.
- No personal data is scraped; all fields are business-level public details.
- Legal review recommended before GA if scope expands beyond MVP fields.

7.5 Data Access Constraints

Constraint Type	Rule	Applies To
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RBAC	Dealer Admin: view and confirm prefilled details; may provide alternate URL. IM: view read-only scrape preview; cannot confirm or save	DWS-SI-00 (IM), DWS-SI-04/DWS-SI-06 (Dealer Admin)
PII Handling	Scraped fields are business-level public details — no PII masking required	All sub-intents
Cross-Dealer Isolation	Scrape cache keyed by (dealership_id, session_id) — no cross-dealer data leakage	DWS-SI-01 through DWS-SI-06
Data Residency	Scraped data cached and stored in US region only for MVP	DWS-SI-02, DWS-SI-03

7.6 Known Risks

Risk ID	Description	Likelihood	Impact	Mitigation
R-001	Dealer website not publicly accessible, behind auth, or has no contact page	High	Low	Silent fallback to TOC-prefilled form (DWS-SI-05)
R-002	Scraped values are wrong or outdated	Medium	Medium	Confidence gate (≥ 0.80); Dealer Admin

				always confirms
R-003	Dealer website blocks scraping via robots.txt	Medium	Low	robots.txt check before scraping; silent skip
R-004	Scrape >5s blocks screen load	Low	High	Hard 5s timeout; async execution; screen loads immediately
R-005	Dealer has no website (small/rural dealers)	High	None	No-website is first-class fallback; no degraded UI
R-006	Dealer Admin provides wrong URL intentionally or accidentally	Low–Medium	High	DWS-SI-08 block-list + fuzzy match (≥ 0.70)
R-007	Dealer Admin provides group HQ URL instead of rooftop URL	Medium	Medium	Fuzzy match; borderline (0.70–0.85) → confirmation prompt

7.10 Financial Controls

N/A — this intent does not handle financial transactions. Dealer website scraping is a data pre-fill feature; no monetary values, GL coding, or financial reconciliation are involved.

§8. Rendering Declaration

8.1 Rendering Mode

Response Type	Preferred Mode	Required Components
Prefilled fields on Dealer Details Review screen	agui (inline form, pre-populated, editable)	Prefill indicator label; confidence badge; editable inputs; confirm/submit button
Scrape failure message	basic (inline advisory, non-blocking)	"We couldn't find your details automatically — please fill in below"
Manual re-scrape trigger	basic (status feedback)	Inline spinner; update fields when complete

8.2 Orchestration Handoffs

Handoff Trigger	Target	Context Passed	Handoff Type
Dealer Admin confirms dealer details	TAO Configure Phase — next step	Confirmed fields (name, address, phone, website_url), dealership_id, session_id	Blocking — Dealer Admin must confirm before TAO advances

§9. Test Scenarios

9.1 Happy Path Scenarios

#	Trigger	Agent Does	Screen Result
1	IM opens provisioning queue; dealer has website URL in TOC record	DWS-SI-00 (TOC sync) → DWS-SI-00b (proactive scrape) → comparison map	IM sees comparison preview: on-file values vs. website values per field

2	Dealer Admin opens screen; TOC sync complete; scrape result cached from IM pre-scrape	DWS-SI-01a → DWS-SI-03 → DWS-SI-04	Dealer Admin sees prefilled fields + diff rows for changed fields; accepts/rejects per field; clicks Confirm
3	Dealer Admin opens screen; small dealer, no website URL	DWS-SI-01a → DWS-SI-01b (no URL) → DWS-SI-05 → contract-only view	All fields populated from TOC; no comparison rows; no website mention
4	Scrape finds same values as TOC on all fields	DWS-SI-03 classifies all as Match	All fields pre-populated with "✓ Matches your website" indicator; no diff rows
5	Dealer Admin provides alternate URL; fuzzy match 0.82 (borderline)	DWS-SI-08 shows confirmation prompt	"We found [name] on this website — is this your dealership? [Yes] [No]"

9.2 Edge Case Scenarios

#	Condition	Agent Does	Screen Result
1	dealer_website_url is null (no website)	DWS-SI-05 → contract-only view	All fields from TOC; no comparison rows
2	Dealer website returns 403 / behind auth	Scrape fails → DWS-SI-05	Same as no-website case

3	Scraped phone confidence 0.62 (below 0.80)	DWS-SI-03 suppresses phone	Phone comparison row not shown
4	Scrape times out at 5s	DWS-SI-05 → contract-only view	Screen loads immediately; no spinner; no error
5	Dealer Admin provides " www.google.com " (block-list)	DWS-SI-08 rejects immediately	"This doesn't look like a dealership website."
6	Scraped name similarity 0.21 vs. contract name	DWS-SI-08 rejects post-scrape	"The website you provided doesn't appear to be for [Dealership Name]."
7	TOC sync (DWS-SI-00) not yet complete when Dealer Admin opens screen	DWS-SI-01a blocked	Screen shows loading state; Dealer Admin cannot proceed

9.3 Edge Case Catalog

ID	Condition	Expected Behavior	Severity
EC-001	TOC sync not complete when Dealer Admin opens screen	Show loading state; block Dealer Admin; notify IM if delayed	Blocking
EC-002	dealer_website_url is null	Skip scraping; render contract-only view	Non-blocking

EC-003	Dealer website behind auth / scrape fails	Scraping skipped; contract-only view	Non-blocking
EC-004	robots.txt disallows scraping	Skip silently; contract-only view	Safety Gate
EC-005	All scraped fields match on-file data	Suppress diff rows; show "✓ Matches your website" per field	Non-blocking
EC-006	User-provided URL on domain block-list	Reject immediately	Blocking
EC-007	Scraped business name similarity < 0.70	Reject; show rejection message	Blocking
EC-008	Borderline match (0.70–0.85)	Show confirmation prompt	Warning
EC-009	Dealer Admin submits with required fields empty	Block submit; validation error	Blocking
EC-010	Dealer Admin re-triggers scrape within 5-minute debounce	Reject; "Please wait before retrying"	Non-blocking

9.4 Golden Test Cases

GTC#	Input	Expected Output	Pass Criteria

GTC-001	Dealer with website; address differs from TOC record	Address diff row shown; Dealer Admin confirms website version	Correct diffs surfaced; no false positives
GTC-002	Small dealer; dealer_website_url=null	All 5 fields prefilled from TOC; zero comparison rows	Screen fully prefilled; zero blank fields
GTC-003	Website response time 6s (timeout)	Screen loads from TOC immediately; no comparison rows	Screen visible in < 500ms; no error
GTC-004	Dealer Admin provides " www.google.com " (block-list)	Immediate rejection; contract prefill unaffected	Immediate rejection
GTC-005	Dealer Admin provides competitor URL (scraped name similarity 0.08)	Post-scrape rejection; contract prefill unaffected	Rejection message shown
GTC-006	trigger_surface="im_provisioning"; 1 address diff found	IM sees diff inline; no save possible	IM read-only view works correctly

9.5 Quality Dimensions

Dimension	Target	Regression Threshold	Eval Method
Accuracy	= 95%	< 90% blocks	Deterministic + LLM-as-judge
Completeness	= 90%	< 85% blocks	Schema validation

Tone	= 4/5	< 3/5 flags	LLM-as-judge
Safety	100%	< 100% blocks	Deterministic + red-team

9.6 Failure Mode Coverage

Failure Type	Applicable?	Mitigation
Logic	Y	Confidence scoring + field-type validation
Integration	Y	5s timeout, robots.txt check, MIME type guard
Behavioral	N	No LLM reasoning in happy path
Quality	Y	Confidence threshold; Dealer Admin always confirms
Drift	Y	Monitor scrape success rates; alert if < 60%
Context	N	Intent scoped to single screen
Safety	Y	Read-only HTTP GET; no cookie/session storage
Coordination	N	Operates independently

§10. Success Metrics & Evaluation

10.1 Business KPIs

KPI	Current Baseline	Target	Timeline
Blank-field rate at screen open	~100% (all fields blank today)	0%	At TAO V1 launch
Dealer Details screen completion time	TBD — measure at TAO V1 launch	Reduce $\geq 40\%$ vs. baseline	30 days post-launch
Dealer Admin field correction rate	N/A	$< 15\%$ of prefilled fields corrected	30 days post-launch
Website diff acceptance rate	N/A	= 50% (Dealer Admin accepts website value when shown a diff)	30 days post-launch
Form validation error rate	TBD — measure at TAO V1 launch	Reduce $\geq 60\%$ vs. baseline	30 days post-launch

10.2 Agent Evaluation Criteria

Metric	Target	Regression Threshold	Cadence
Intent match rate	95%	$< 90\%$ triggers review	Weekly
Golden set pass rate	98%	$< 95\%$ blocks release	Per deploy

Invariant violation rate	0%	0% blocks release	Continuous
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10.3 Invariants (Continuous Checks)

Invariant	Description	Enforcement
Layer 1 must precede Layer 2	Screen must never render without Layer 1 (TOC data) present	Hard block — DWS-SI-01a must succeed before DWS-SI-04 renders
Screen is never blank	TOC values fill every field; Layer 2 cannot leave a field emptier than Layer 1 left it	Hard block
Read-only scrape	Scraper NEVER submits forms, writes cookies, or authenticates	Hard block — reject if non-GET HTTP method used
Dealer Admin confirmation required	No value persisted without explicit Dealer Admin submit	Hard block — DWS-SI-06 must receive a Dealer Admin-triggered event
Dealer isolation	Results for dealership_id=X never shown to session for dealership_id=Y	Hard block — validate session dealership_id matches cached result
URL validation before user-provided scrape	DWS-SI-02 must never run on dealer_provided_url unless DWS-SI-08 passed it	Hard block
IM cannot save	DWS-SI-06 rejects calls where	Hard block

	trigger_surface="im_provisioning"	
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10.4 Related Intents & Dependencies

Upstream: TAO-CONTRACT-001 (dealer_website_url must be captured at contract signature); TAO-COA-001 (same Configure Phase, no direct dependency)

Downstream: TAO Configure Phase next step — DWS-001 completion unblocks the next step in the Configure Phase sequence.

§11. Follow-Up Patterns

Pattern	Trigger	Agent Response
Dealer Admin edits a prefilled field	Dealer Admin changes a prefilled value	Accept silently; edited value replaces prefilled value — Dealer Admin is always the source of truth
Dealer Admin requests re-scrape	Dealer Admin triggers /prefill-dealer-details	Check debounce (5 min); if allowed, re-run DWS-SI-02→DWS-SI-04; if within debounce, show "Please wait before retrying"
Prefill partially succeeds	DWS-SI-03 returns partial field map	Prefill found fields; leave unfound fields blank with placeholder text; no partial-failure message

§12. Technical Approach

Architecture — Two-Layer Model

Data flow: Salesforce → TOC → TAO

Layer 1 — Internal Data Sync (baseline, always runs)

1. **TOC Sync Service** — At IM provisioning stage, reads the dealer record from TOC. TOC already holds the Salesforce-synced contract data (legal name, address, phone, website URL) plus any provisioning-stage updates. TAO does not connect to Salesforce directly.
 2. **TAO Dealer Record** — Stores the TOC-sourced dealer record. Dealer Admin screen reads from here. Always populated before the Dealer Admin journey begins. This is a **blocking dependency** for the Dealer Admin screen load.
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Layer 2 — Website Scraping + Comparison (enrichment, additive)

3. **Scrape Trigger** — Resolves dealer website URL from TAO dealer record. If no URL → Layer 2 skipped entirely; Layer 1 data shown alone.
 4. **Web Scraper Worker** — HTTP GET to target URL. Enforces: 5s timeout, robots.txt check, MIME type guard (HTML only), read-only GET only. On failure → Layer 1 shown alone; no error to Dealer Admin.
 5. **Field Extractor** — Parses raw HTML; extracts dealership name, address, phone, website URL. Assigns confidence score (0.0–1.0) per field. Fields below 0.80 suppressed.
 6. **Field Comparator** — Classifies each field as: match / different / new / missing. Returns structured comparison map.
 7. **Comparison View Renderer** — Injects comparison map into screen. TOC values always shown as baseline. Diff/new rows shown for Dealer Admin decision. Match fields show quiet checkmark.
 8. **Session Cache** — Caches comparison result by (dealership_id, session_id). IM pre-scrape result reused on Dealer Admin screen load if not stale.
 9. **TAO Dealer Record Write** — On Dealer Admin submit, confirmed values (TOC baseline + any website values accepted) persisted via existing dealer details save endpoint.
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URL Validation (Dealer Admin-provided alternate URLs)

10. **URL Validator** — Domain block-list check (immediate). If passes → scrape → business name fuzzy-match against contract name. < 0.70 → reject. 0.70–0.85 → confirmation prompt. >= 0.85 → accept silently.
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Non-Functional Requirements

- **Screen load latency:** Dealer Admin screen must render from Layer 1 data in < 500ms. Layer 2 runs asynchronously; never blocks the screen.
- **Layer 1 is blocking:** Dealer Admin cannot open the screen until TOC sync (DWS-SI-00) is complete.
- **Layer 2 is always additive:** Any Layer 2 failure falls back silently to Layer 1 data. No Dealer Admin-visible error.
- **Rate limiting:** Max 1 scrape per (dealership_id, session). Max 1 user-triggered re-scrape per 5 minutes.
- **Security:** Read-only HTTP GET only. No auth. No cookies. No form submissions.

§13. Rollout Plan

Rollout Target: TAO V1

Ships with TAO V1. Controlled by feature flag `dws_prefill_enabled` for rapid kill-switch.

Gate	Criteria	Owner
Pre-launch	Zero P0/P1 bugs in GTC-001 through GTC-006	Engineering
Pre-launch	DWS-SI-08 domain block-list reviewed and approved	PM
Pre-launch	Fuzzy match thresholds (0.70/0.85) validated on sample of real dealer names	Engineering + PM
Pre-launch	IM provisioning view updated with scraped preview column	Engineering
Pre-launch	"Use a different website" UI component built and tested	Engineering + UX
Post-launch monitoring	Scrape success rate >= 50%; timeout rate	PM

	<= 10%; Dealer Admin edit rate on prefilled fields < 25%	
Rollback trigger	P0: any scrape that writes to a dealer website; any prefilled value saved without Dealer Admin confirmation	Engineering on-call

Feature flag: `dws_prefill_enabled` — set to `false` disables all scraping immediately. No data migration needed for rollback.

Appendix A: ISD Versioning

Version	Date	Author	Change Summary
1.0	2026-08-27	Prem Bihari (pbihari@tekion.com)	Initial ISD draft
1.1	2026-08-27	Prem Bihari (pbihari@tekion.com)	Expanded DA abbreviation to Dealer Admin throughout